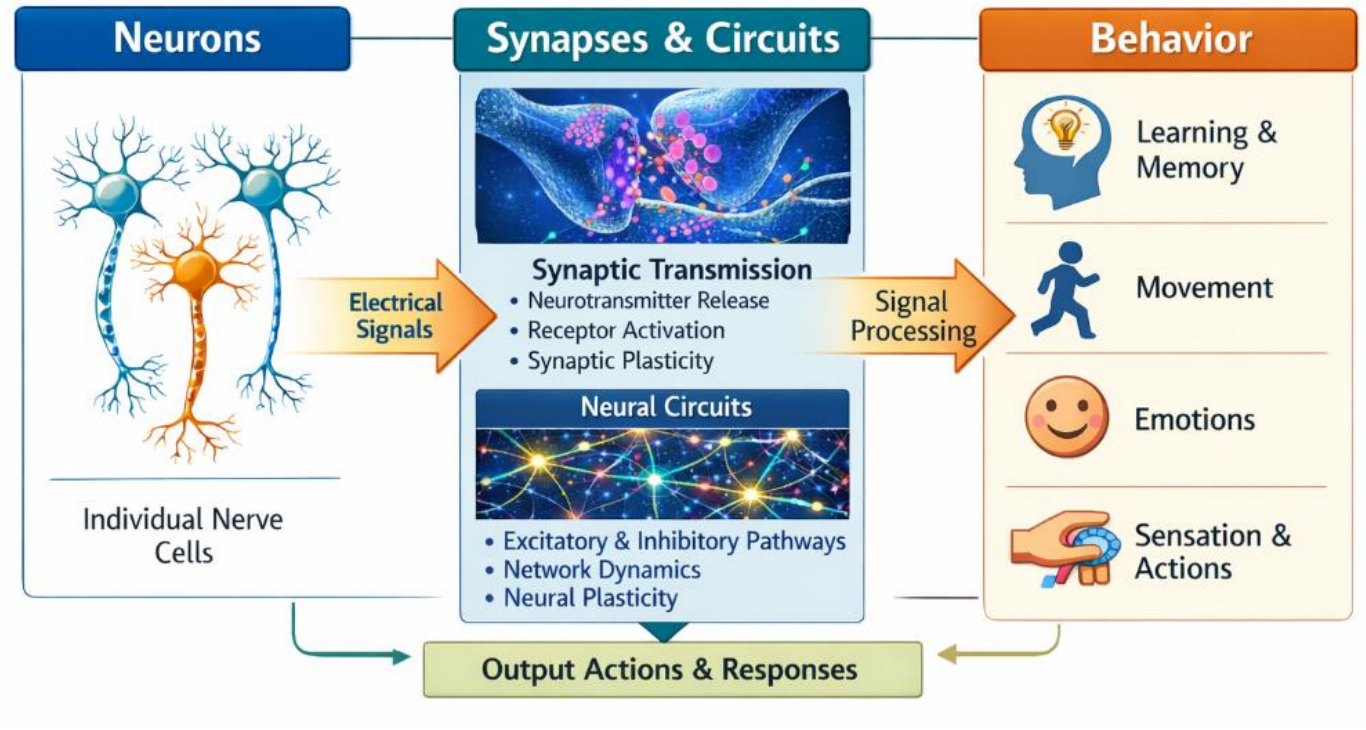
Behavioral Neuroscience

Basic Concepts in Animal Behavior and Neuroethology

Tali Kimchi
tali.kimchi@Weizmann.ac.il

April, 2026

The nervous system evolved to generate behavior



Outline

Defining behavior

What are the different aspects of behavior research (Tinbergen's four questions)?

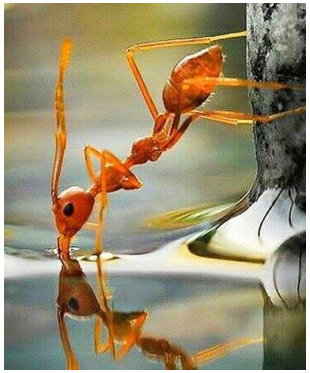
Type of behaviors (innate vs learned)

Neuroethology - definition and examples

What is animal behavior?

The study of how animals interact with their environment, other animals, and themselves

Environment



Other animals



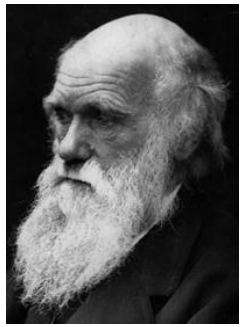


Themselves
(reflex movement)



Everything an animal does

How does behavior evolve? Natural selection



Charles Darwin
(1809-1882)

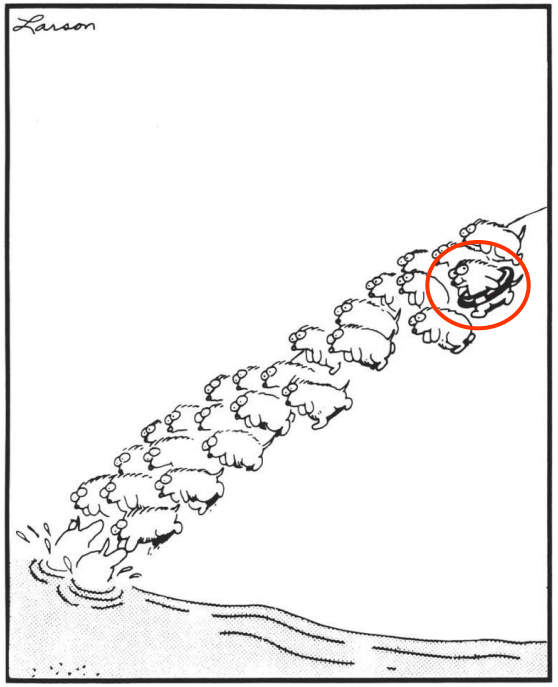
Behavior (like other character) is shaped by natural selection.

1. Variations also exist in behavioral traits

2. Some of these behavioral variations are heritable

3. Certain behavioral variations make individuals better adapted to their environment

4. These individuals are more likely to survive longer and leave more offspring than those with less successful behavioral traits



Evolutionary causes of behavior

In nature, animal behave in ways that maximize their fitness

Fitness = the ability to survive and reproduce



Beneficial behavior





Non-beneficial behavior

Sexual selection of behavioral traits (Darwin 1871)

Traits directly related to reproductive

Traits that are not favored by natural selection







Why doesn't natural selection eliminate costly traits produced by sexual selection?"

Different behaviors serve different evolutionary functions

Foraging: maximize energy gain while minimizing cost
(Optimal foraging theory)

Courtship: maximize reproductive success
Males: facilitate attraction of females
Females: evaluate the quality of potential mates



Tale of the peacock

Why does the peacock have such a beautiful tail? Darwin's theory of sexual selection says that nature's extravagances—like the peacock's tail—are advantages in the mating game.





Decorative plumage,
accounts for 60% of their length

https://www.pbslearningmedia.org/resource/tdc02.sci.life.evo.peacock/tale-of-the-peacock/

Costly Signaling theory (Amotz Zahavi's handicap principle)

Traits that are costly to develop and maintain can act as "honest" signals of a male's high genetic quality





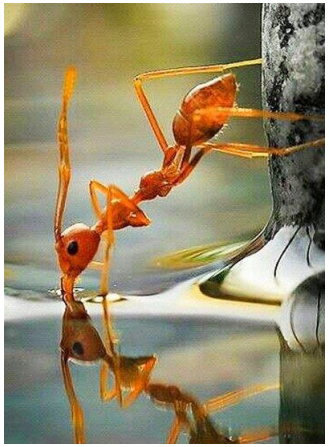
Immediate causes of behavior

Animals behave in response to a stimulus- an external (e.g. odor, sound) or internal (physiological) signal

External signals

Internal signals

Physical stimulus



Social stimulus



Sex hormones
(Testosterone elevation, trigger aggression)

Reproductive stage

Species-specific sensory systems adapted to specific environmental and social stimuli



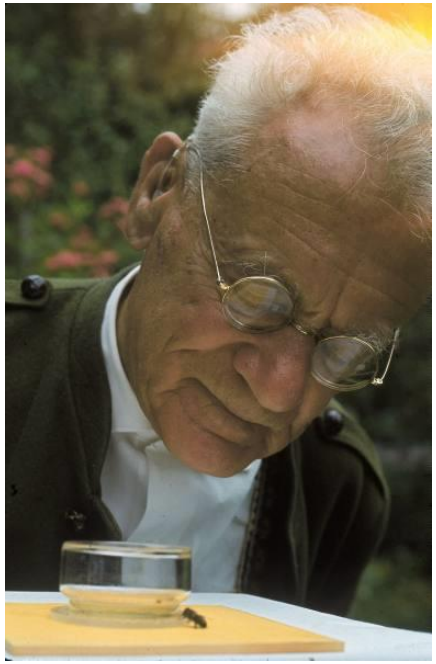


Studying animal behavior

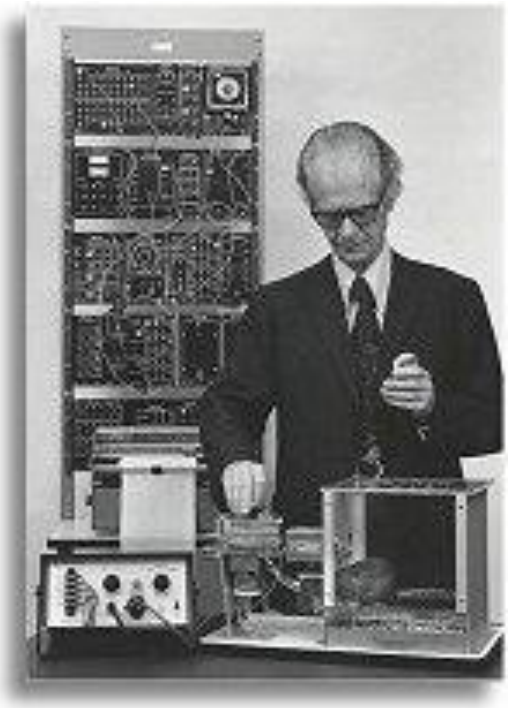
Scientific Method

- Initial Observation(s)
- Formulate a question
- Phrase null and alternate hypotheses
- Plan an experiment that distinguishes between two hypotheses
- Conduct experiment
- Analyze data
- Draw conclusions

Ethologist



Behaviorist



-Ethologists study focus on animal behavior in natural habitats to understand evolutionary, innate, and instinctual patterns, often focusing on survival (Zoology).

-Behaviorists study focus on how animals learn, using experimental settings to study conditioned responses, rewards, and punishments (psychology).

Ethology

The study of naturalistic animal behavior in the field and its relationship to its evolutionary origins

Ethological research starts with observations of the animal's natural behavior in its natural habitat





The key principle is that context defines function

Ethology guidelines

If you want to understand anything about the biology of behavior:

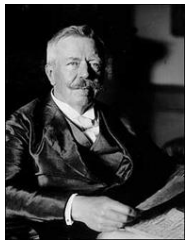
Assume that you are going to have to translate a language

Speak to the animal in its own “language”

-thinking how the world looks through the animal’s senses

(**the animal “Umwelt”)**

The animal "Umwelt"



Jakob von Uexküll (1864-1944)

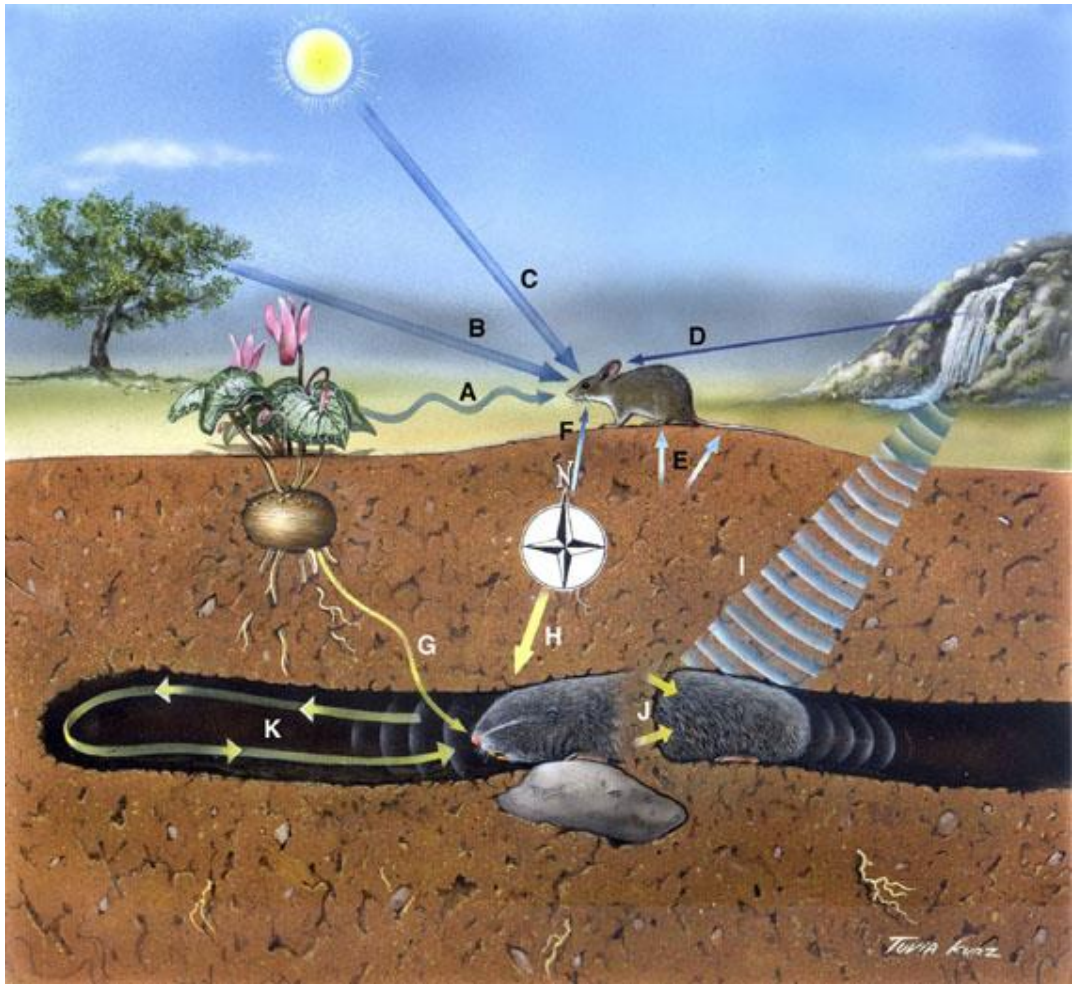
Umwelt = The organism's model of the world

How an animal perceives the world and the actions (behaviors) that are performed by each species.



Human: 400-780 nanometers

Bee (UV vision): **300**-560 nanometers


A
B
C
D
E
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N
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H
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J
K





Ethologist observe the natural animal behavior and choose the appropriate animal model for the research question



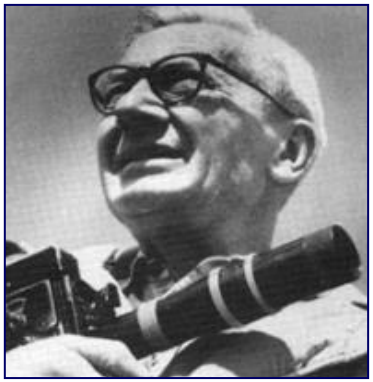


Three pioneering observes of animal behavior (Ethology)

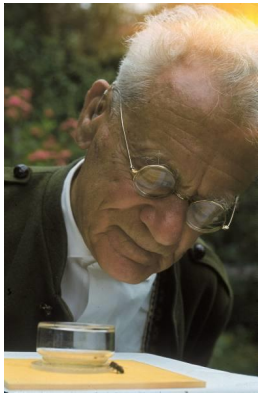
1973 the Nobel Prize for Physiology or Medicine



Konrad Lorenz
(1903-1989)



Nikolaas (Niko) Tinbergen
(1907-1988)



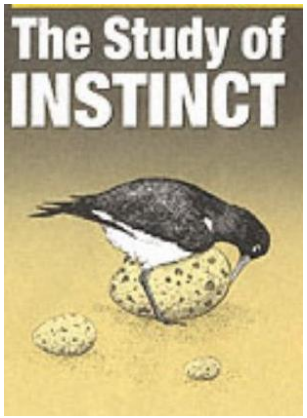
Karl von Frisch
(1886-1982)

Experts in animal behavior who, through extensive field observation and experience, identified fundamental patterns and motivations in the behavior of animals

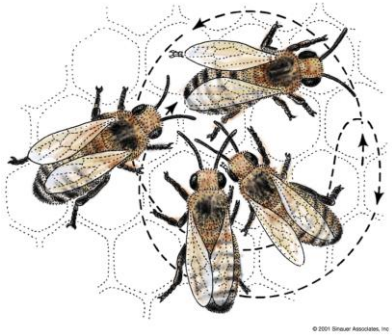
Three pioneering observes of animal behavior (Ethology)



Konrad Lorenz:
Imprinting



Niko Tinbergen:
Fixed action pattern



Karl von Frisch:
Honeybee communication

Their main discoveries were made on insects, fishes and birds, but the basal principles have proved to be applicable to all mammals

How do we study and understand behavior?

Asking *how* and *why* animals interact with each other and their environment

Observation: define the behavior

How does it work?

How did it develop?

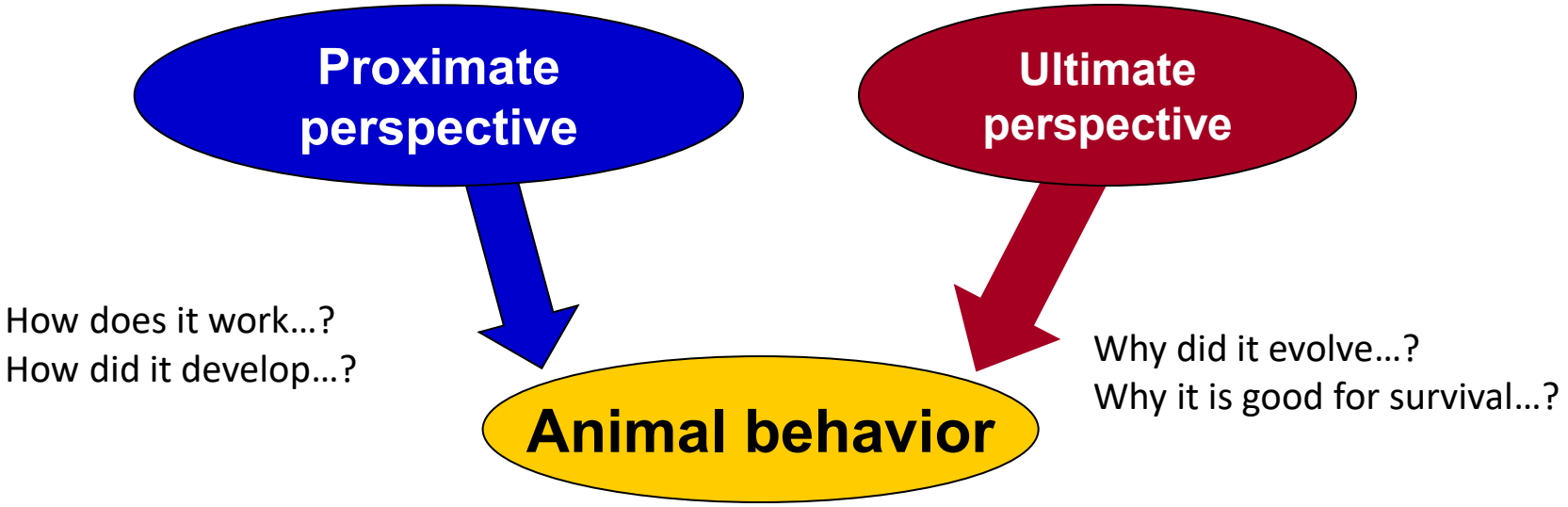
Why did it evolve?

Why it is beneficial for survival?

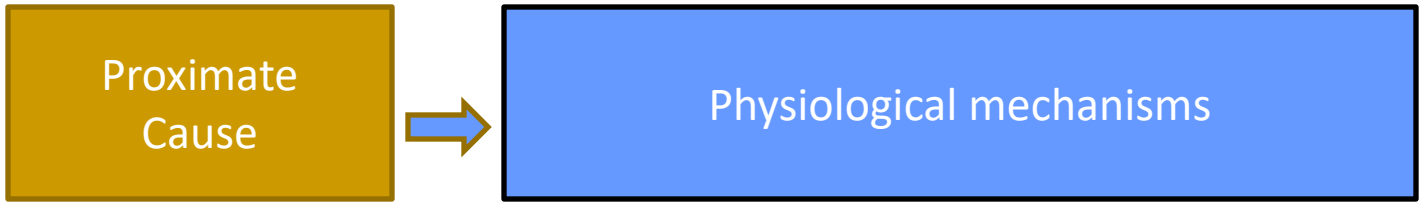


Nikolaas Tinbergen

The 4 questions asked in ethology (Niko Tinbergen)



	Short-term	Long-term
How?	Mechanism	Development
Why?	Function	Evolution



Studying the mechanisms of the immediate causation and development of the behavior



Studying the adaptive significance and evolution for survival and reproduction

Lion Intimidation

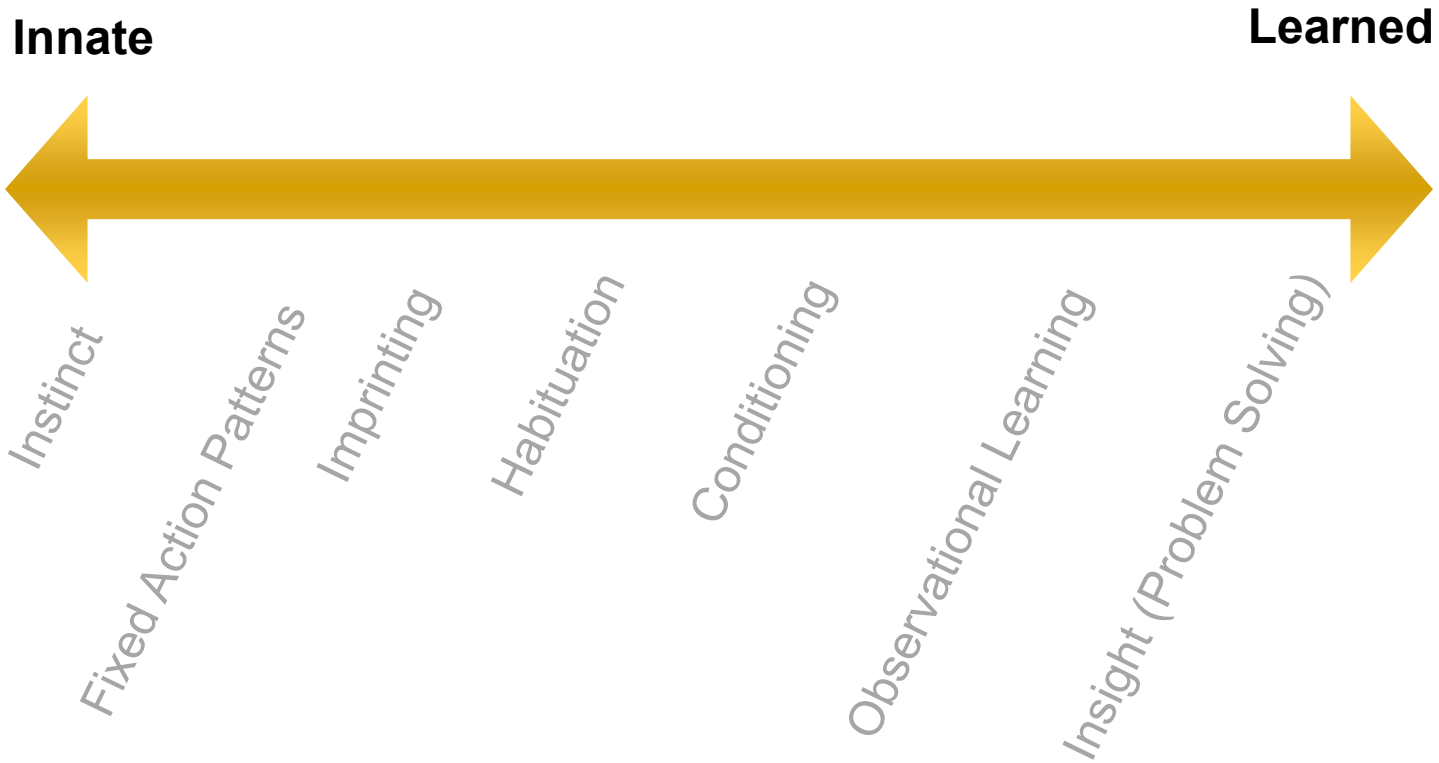
Young lions have to leave their pride and take over other prides



Lion intimidation	*Short-term*	*Long-term*
How?		
Why?		

	Short-term	Long-term
How?	Mechanism	Development
Why?	Function	Evolution

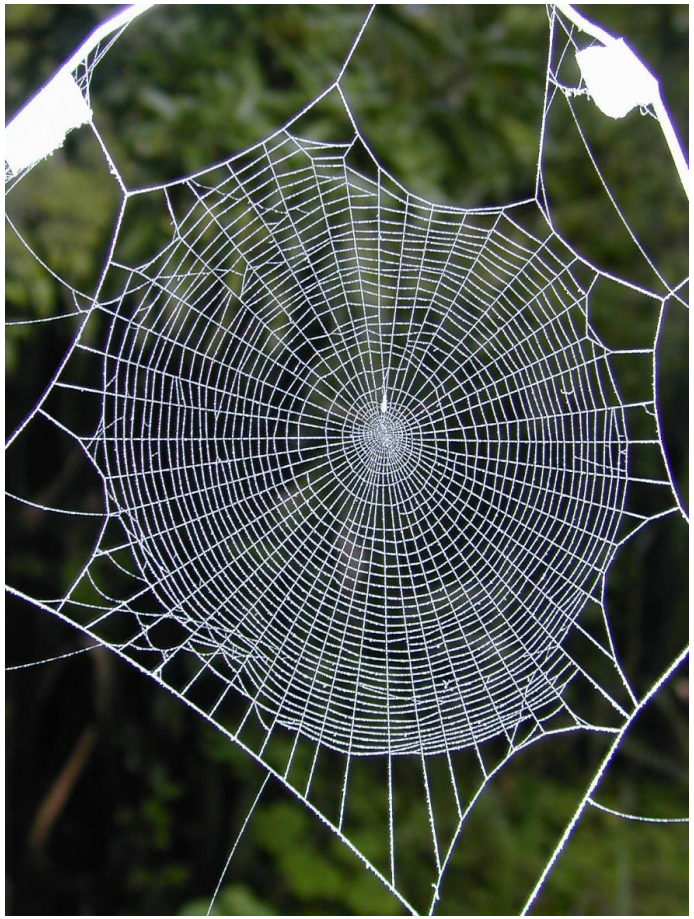
Types of behavior

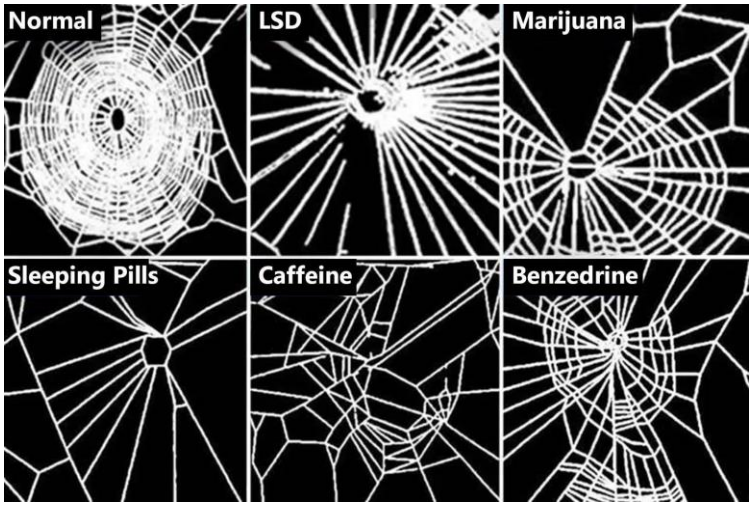


Innate behavior

- First time performance is completely functional
- Animals don’t have to witness the behavior (inborn)
- Uniform, stereotyped
- Triggered by a simple sign stimulus (sensory releaser)
- Has a strong genetic (inherited) basis: controlled by pre-programmed neural circuitries

Innate behavior





https://rarehistoricalphotos.com/nasa-spiders-drugs-experiment/

Innate behavior that is shaped by learning with experience





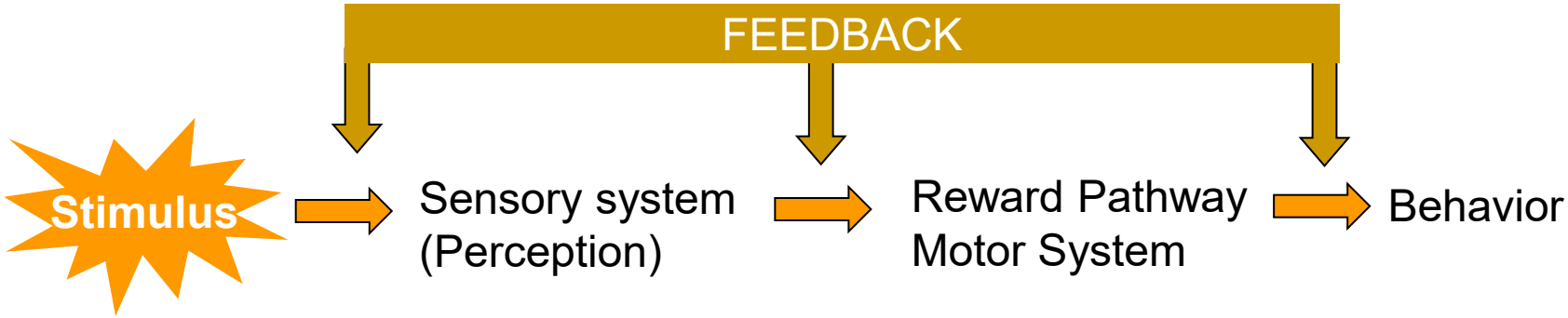
Innate behavior that is shaped by learning with experience





Learned behavior

- Behavior that is modified by experience (trial-and-error pattern)
- Flexible. The phenotype changes with time/experience
- May overrides/refined innately programmed behaviors



How to crack a nut?



Pick the right stones

Learn to handle one object

Combining 2 objects

Place the 2 objects in the right order

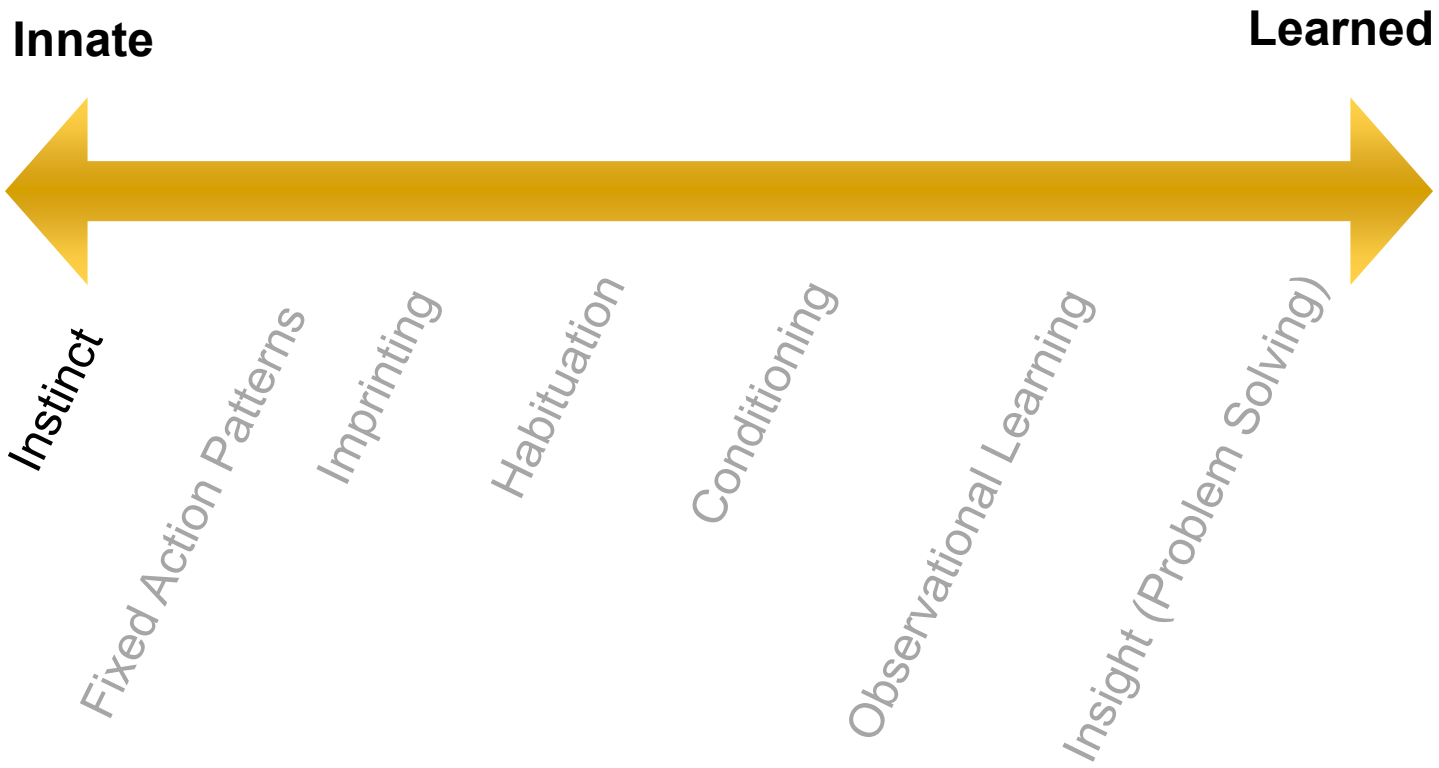
Place all 3 objects in the right order

Hit the right power to crack the nut

Innate or Learned?



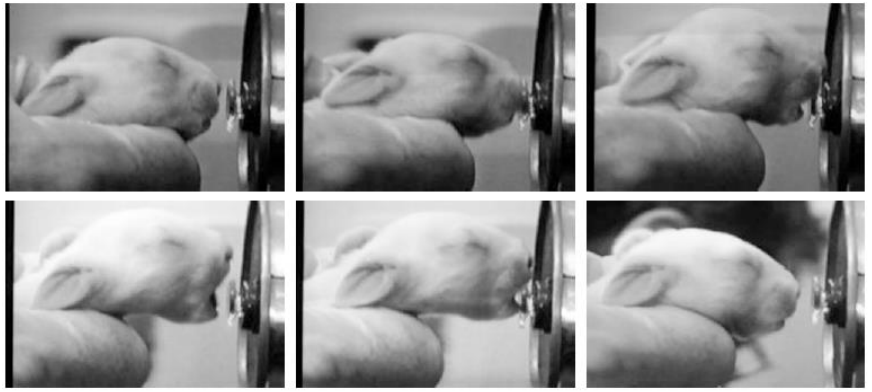
Type of behaviors



Suckling behavior in newborns

Newborn mammals must eat to survive. Thus, precise, genetically defined developmental programs must be executed prenatally to ensure effective nursing at birth





Schaal et al 2003

Newborns tend to suck on anything that touches the roof of their mouth

Grasping behavior in newborns

Involuntary movement where a newborn closes their fingers tightly around an object touching their palm







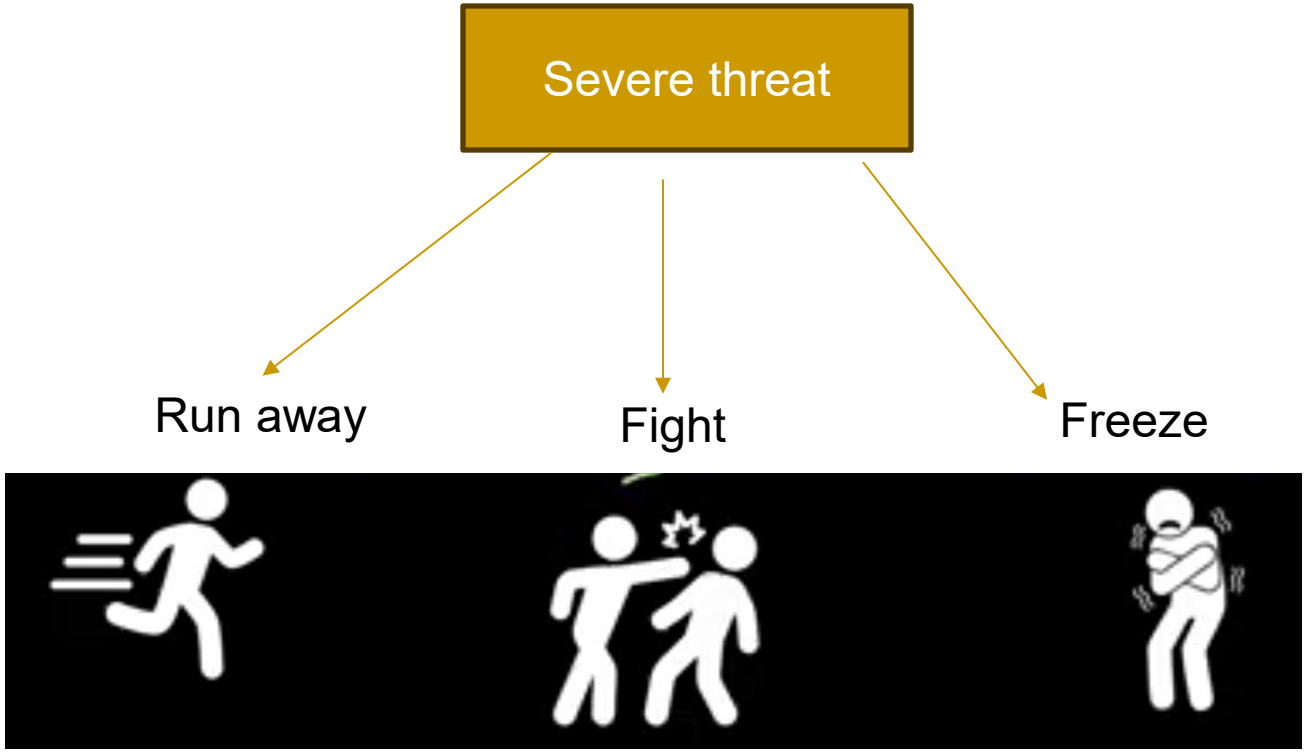
It is a primitive survival instinct that remains strong, in humans, until about 5-6 months, when it transitions to voluntary grasping

Hiding behavior in newborn deer



Newborn deer (fawn) possess a powerful, instinctual hiding behavior known as "freezing," for the first few weeks of life.

Flight or Fight Response (Freeze or Run)

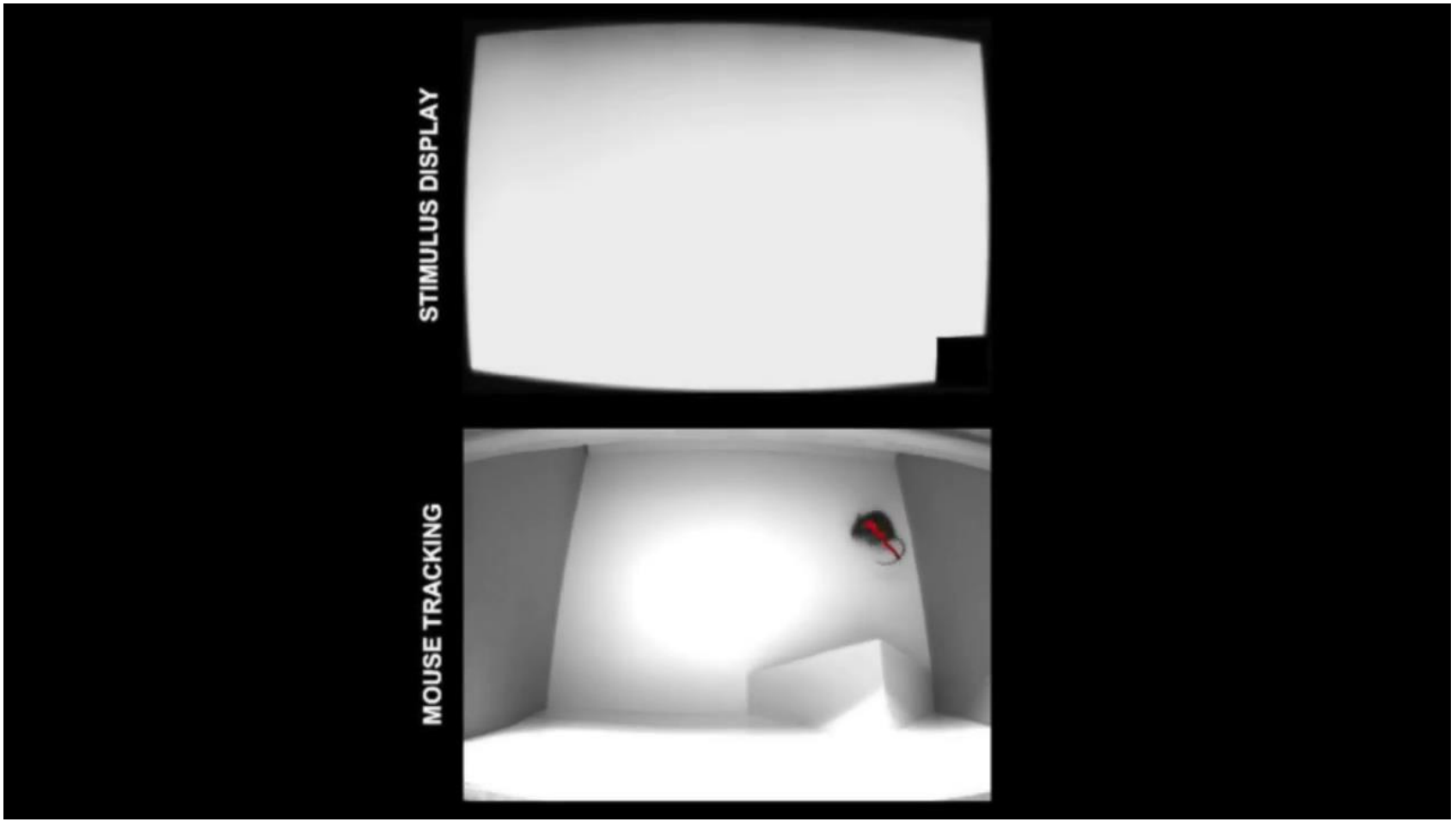


Critical for survival (stimulus-specific, species-specific)

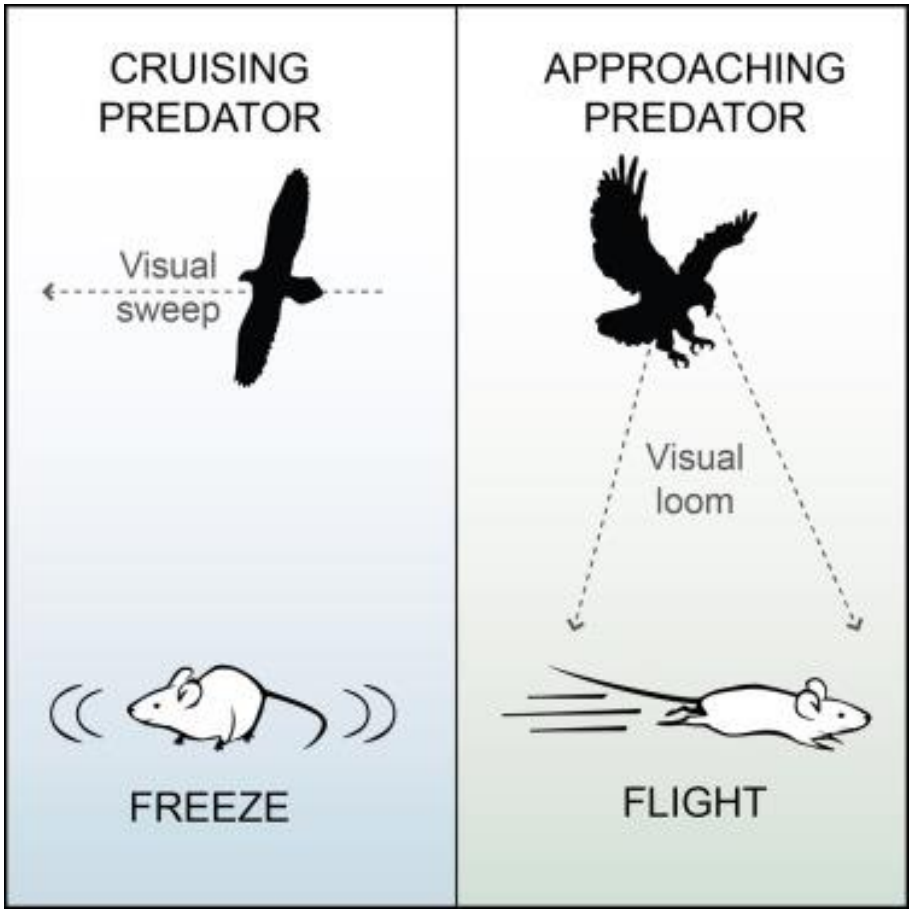
Instinct behaviors: Freeze or flee in wild-caught mice



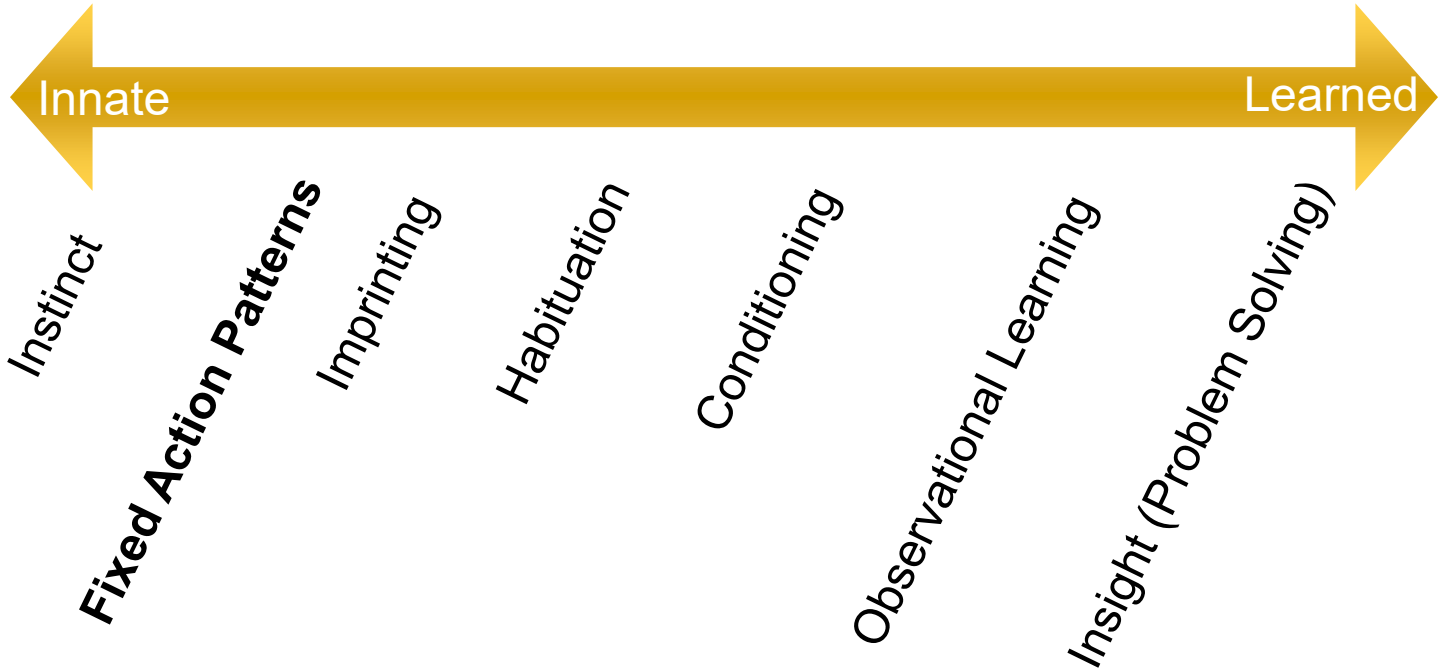
Instinct: Freezing + Fleeing



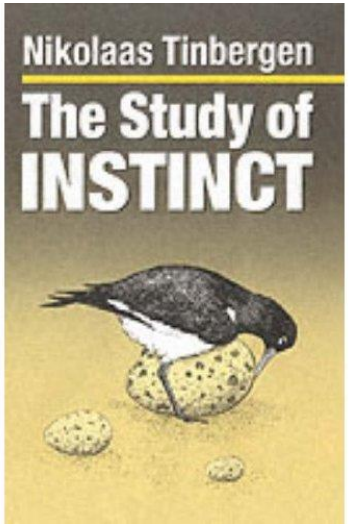
Instinct behavior in the field



Type of behaviors



Fixed Action Pattern (FAP)

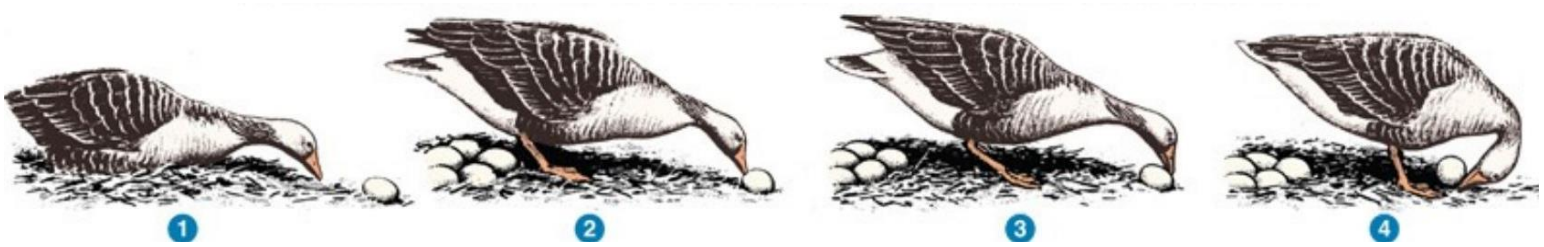




Nikolaas Tinbergen

- Highly stereotypic behavior
- Once begun, the behavior will continue to completion
- Triggered by a sign stimulus (external stimulus)

FAP:
Egg-rolling behavior in greylag goose



- The goose will roll an egg that is outside the nest back into the nest in the same manner every time.

- Each time this action pattern is initiated, it is carried through to completion.

- The goose will do this with any round object placed outside the nest.

FAP:
Egg-rolling behavior in greylag goose



Each time this action pattern is initiated, it is carried through to completion.

FAP:
Egg-rolling behavior in greylag goose



Visual sign stimulus: The goose will do this with any round object placed outside the nest.

Supernormal stimulus

An exaggerated stimulus will elicit a stronger response



FIG. 69. Grey lag goose retrieving egg. After Lorenz and Tinbergen, 1938.

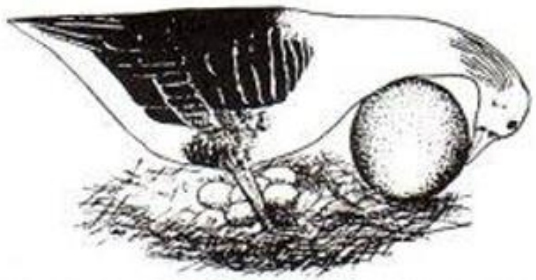
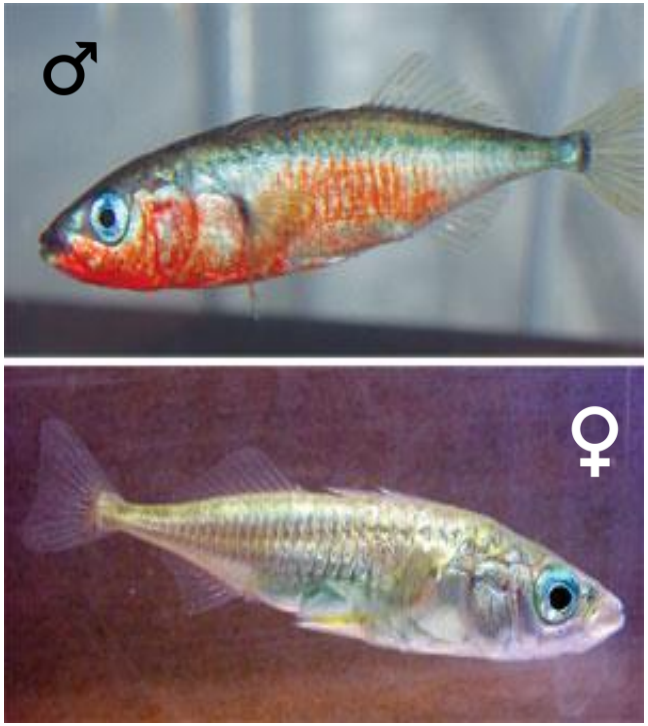
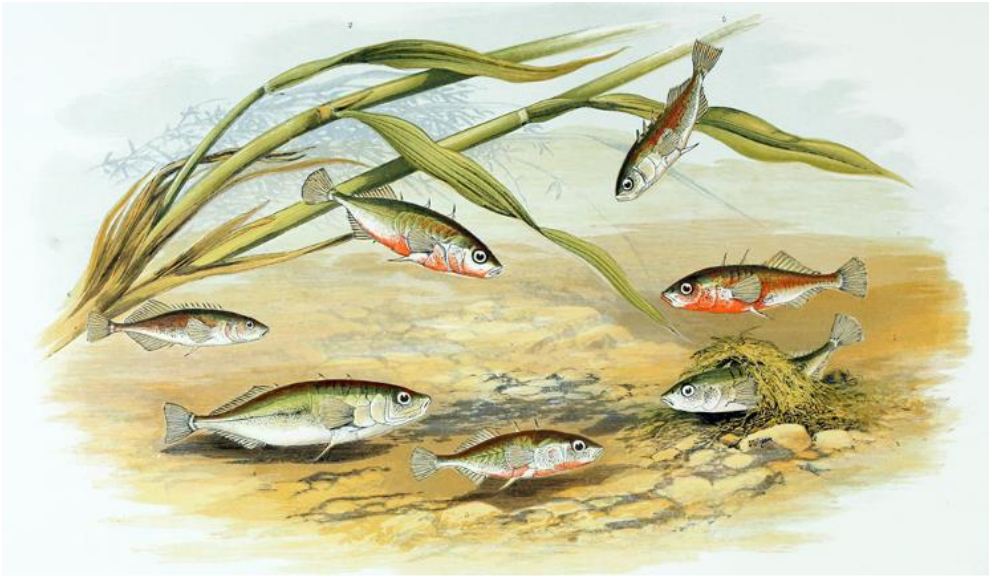


FIG. 70. Grey lag goose attempting to retrieve giant egg. After Lorenz and Tinbergen, 1938.

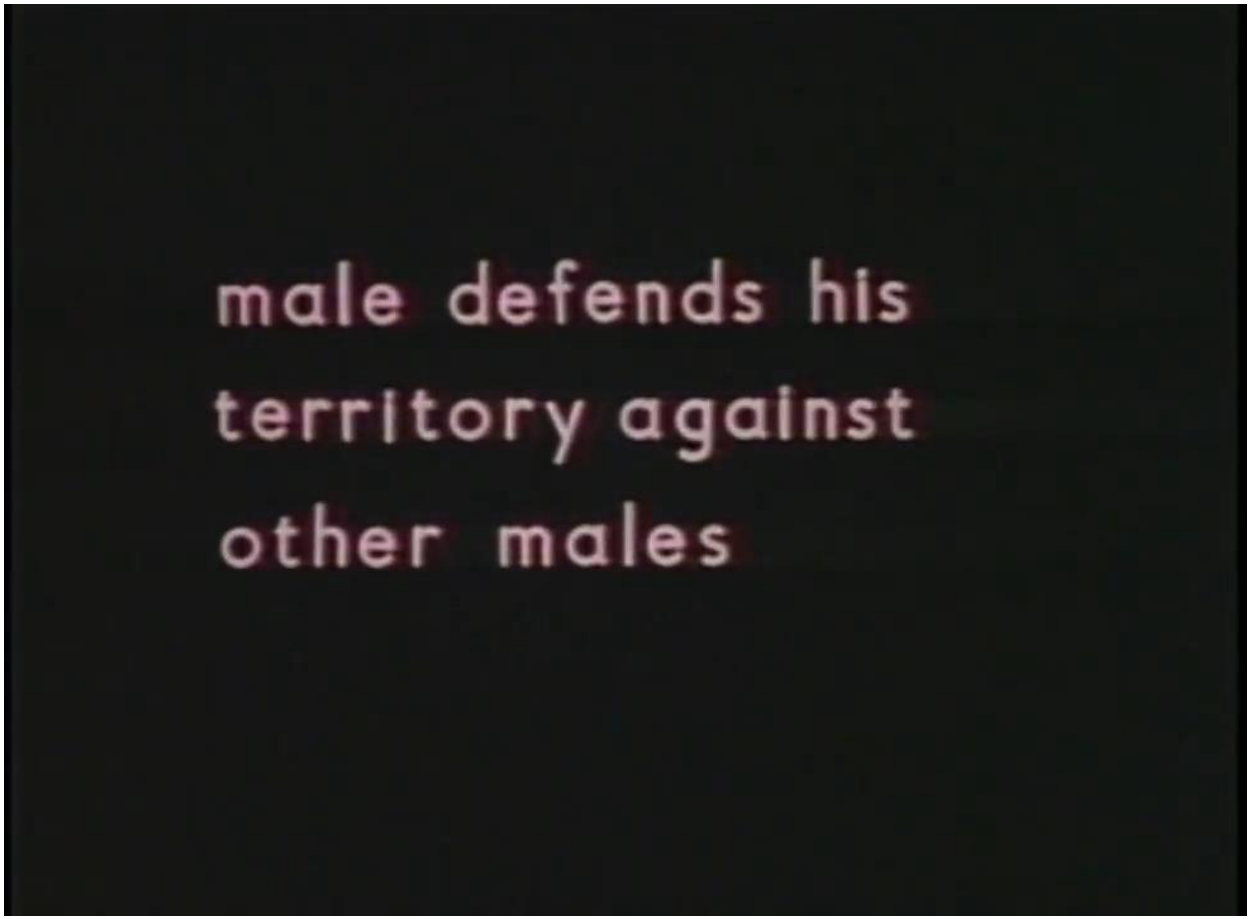
Ultimate causation:
Larger eggs typically produce healthier chicks

FAP:
Reproductive behavior in three-spined stickleback

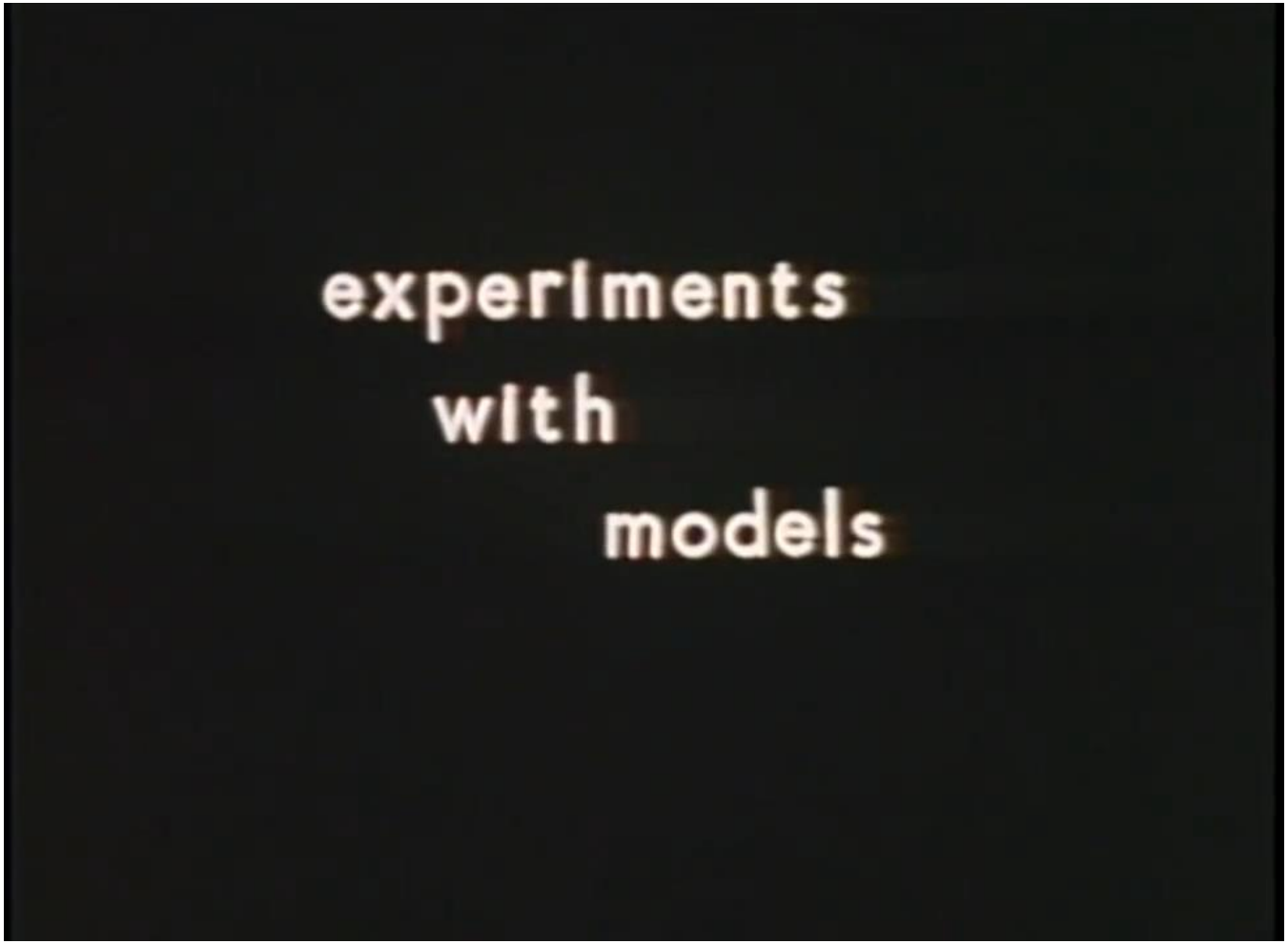




Stickleback – Aggressive behavior; field observation



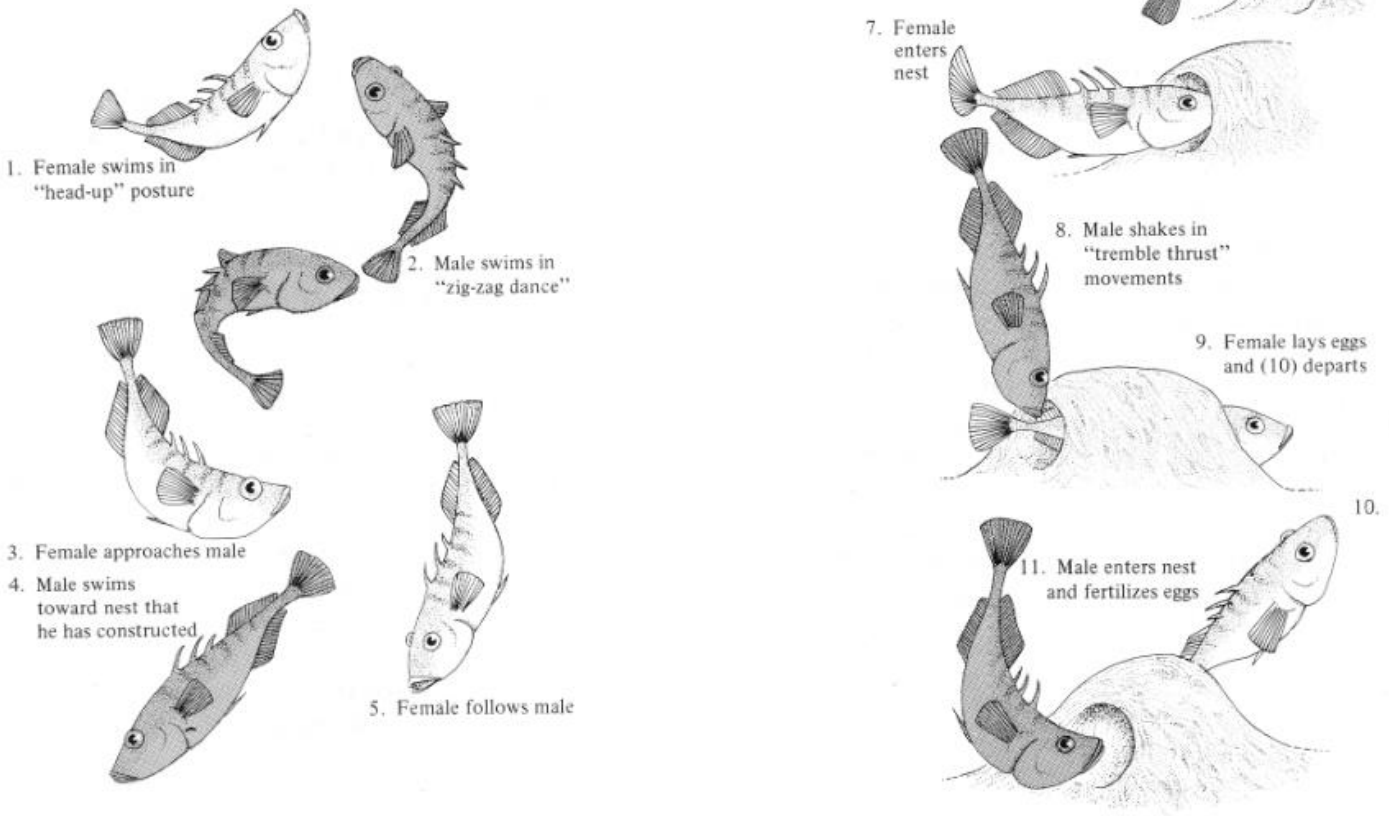
Stickleback – FAP (aggression); lab experiment with models



Signal releaser for aggressive behavior – red color belly

Stickleback – Courtship behavioral repertoire

Fig. 24.2 A courtship sequence between stickleback fishes. Reprinted by permission from Donald D. Ritchie and Robert Carola, *Biology*, © 1979. Reading, Massachusetts: Addison-Wesley (Fig. 28.19).



Stickleback – Courtship behavior; field observation



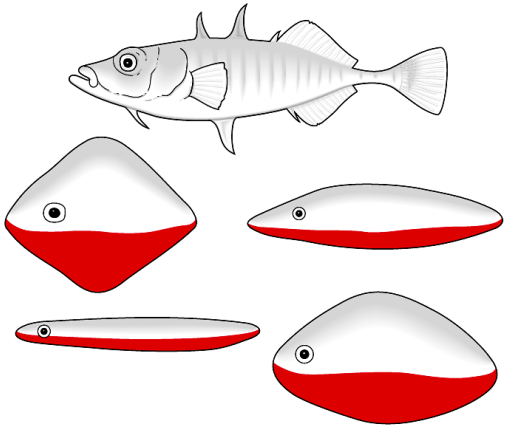
Stickleback- FAP (courtship behavior); lab experiment with models



Signal releaser for courtship behavior– fat object (white swollen belly)

Fixed action pattern in stickleback

- Will attack: as long as a red spot is present on the ventral part of the body (shape is not important)


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- Will court if: white swollen belly



Proximate and ultimate perspectives on FAP in male sticklebacks

BEHAVIOR: A male stickleback fish attacks other male sticklebacks that invade its nesting territory



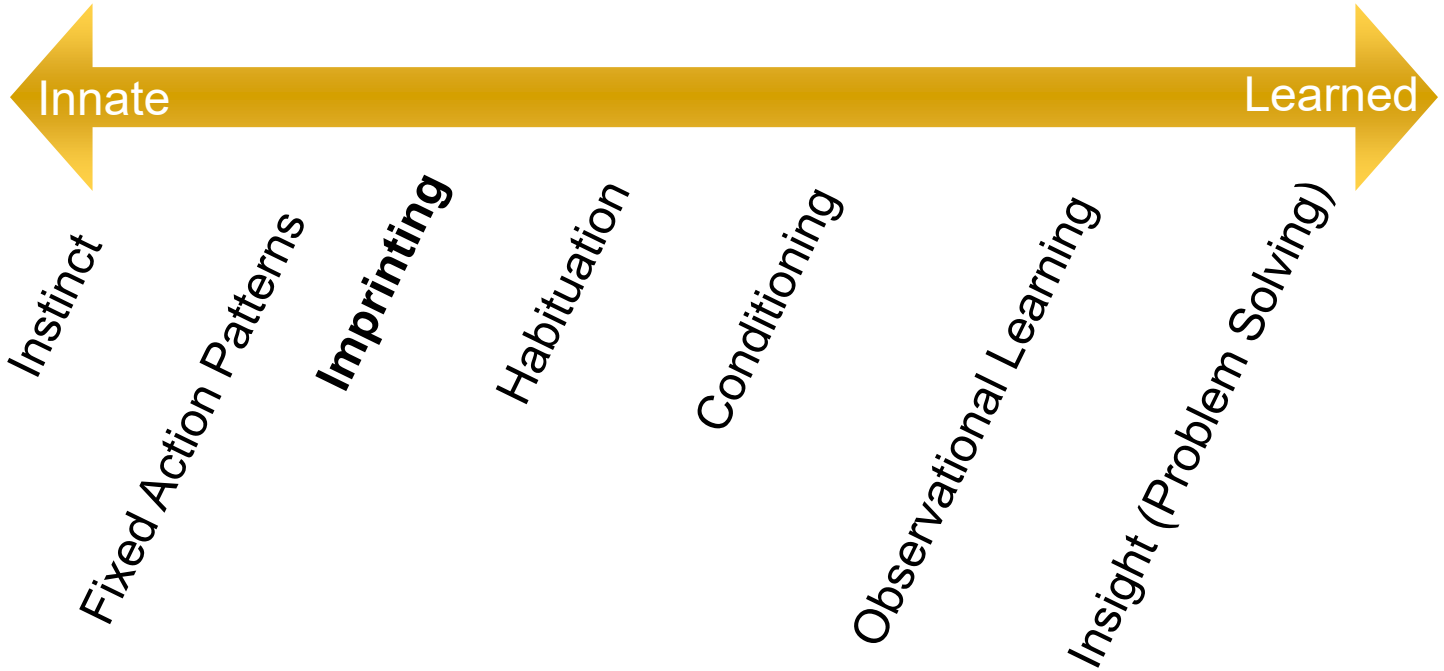
PROXIMATE CAUSE:

The red belly of the intruding male acts as a sign stimulus that releases aggression in a male stickleback

ULTIMATE CAUSE:

By chasing away other male sticklebacks, a male decreases the chance that eggs laid in its nesting territory will be fertilized by another male

Type of behaviors



Imprinting behavior - Konrad Lorenz

- A learned behavioral pattern that is dependent on innate mechanisms
- Learning that occurs during a critical period in the early life of an individual
- Irreversible



Imprinting field observation

Behavioral observation: Geese hatchlings closely follow their mother

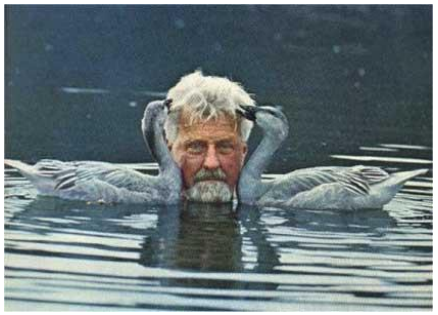
Explanation: Mother-offspring bonding in animals is crucial to safety & development of the offspring

Hypothesis: Geese hatchlings follow the first thing they see moving





Lorenz's imprinting experiment



Experiment: A clutch of goose eggs was divided between the mother goose and an incubator (treated by Lorenz).

Results: Goslings reared by the mother behaved normally and mated with other geese.
-Goslings that spent their first hours of life with Lorenz followed him and even tried to mate with humans.

Conclusions: Goose have no innate sense of "mother" or "gooseness".
-They identify with and respond to the first object with certain characteristics they encounter.
-The ability or tendency to respond is innate.

Learning who is your mother- Imprinting behavior



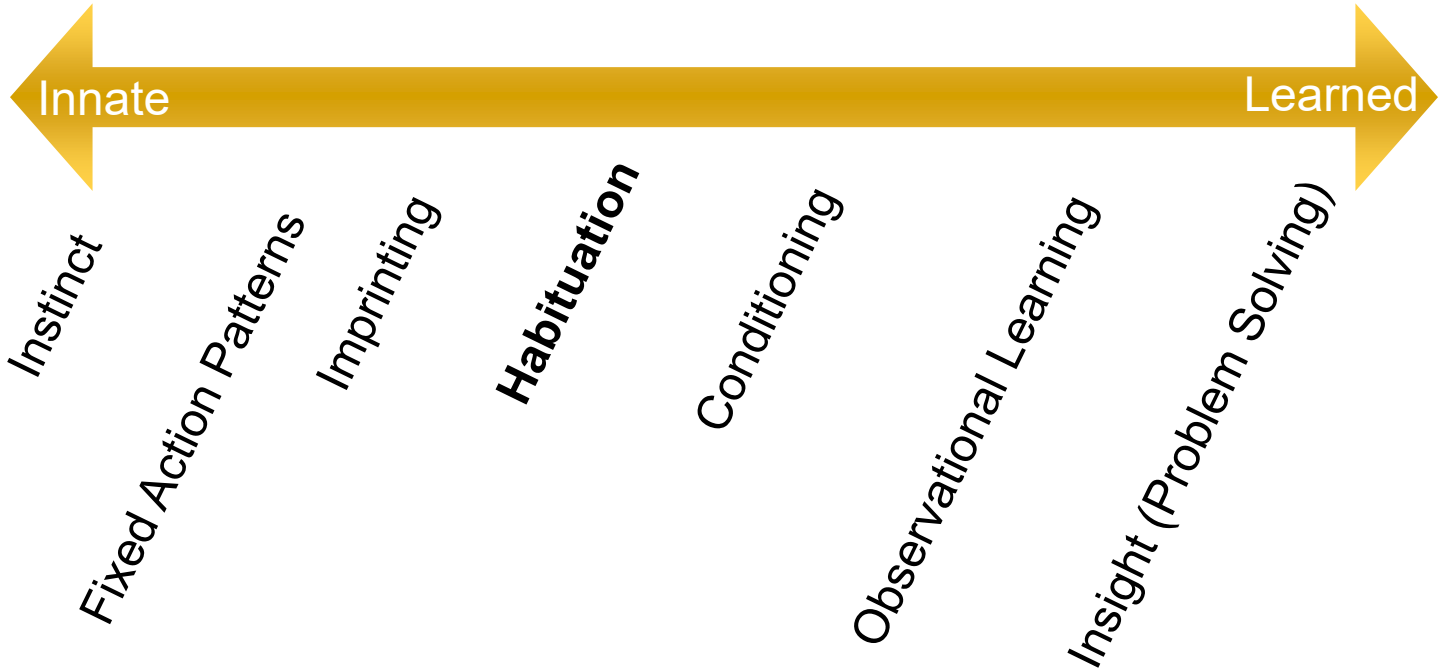
Imprinting for conservation

Conservation biologists have taken advantage of imprinting by young whooping cranes to teach the birds a migration route.



Break

Type of behaviors



Habituation – Reduction of innate responses

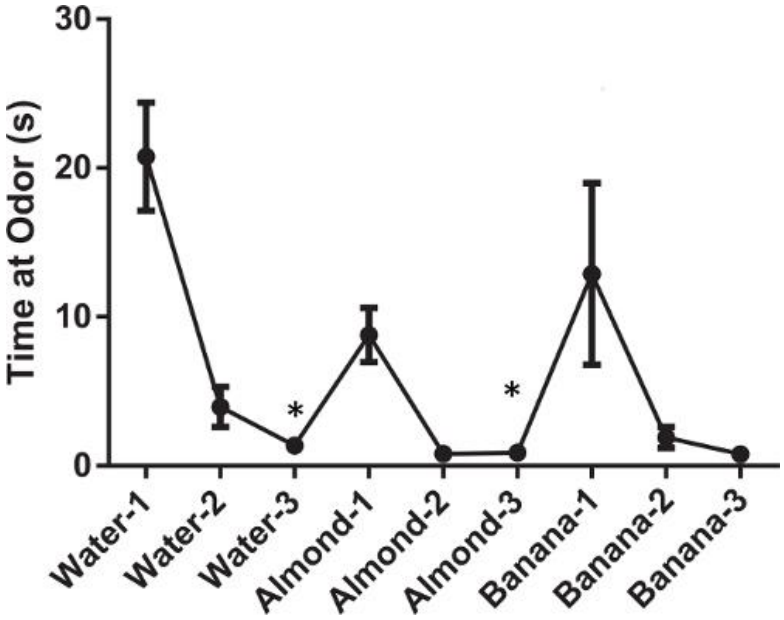
Animal's response to a novel, neutral (not dangerous) stimulus weakens after repeated exposure

Animal learns to ignore harmless stimulus that has no consequence (no apparent significance in its life)

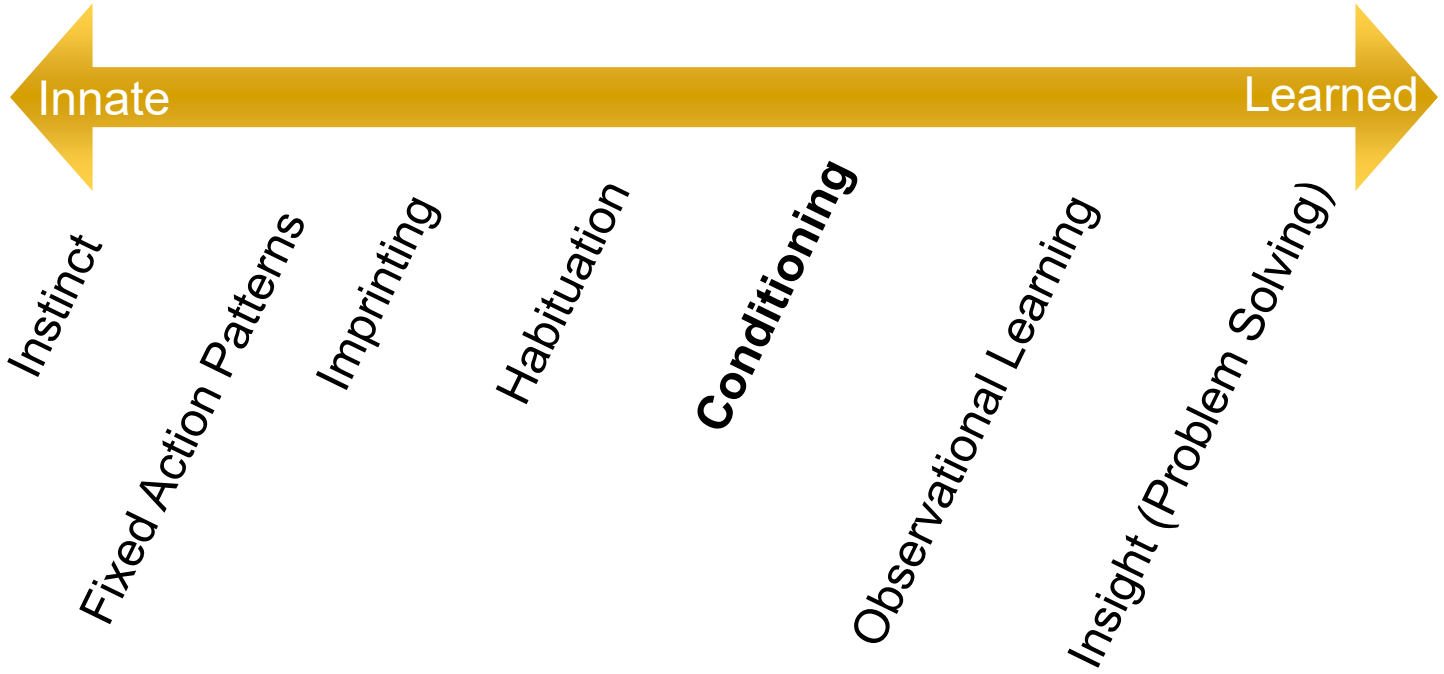


Habituation to neutral odors in mice





Type of behaviors

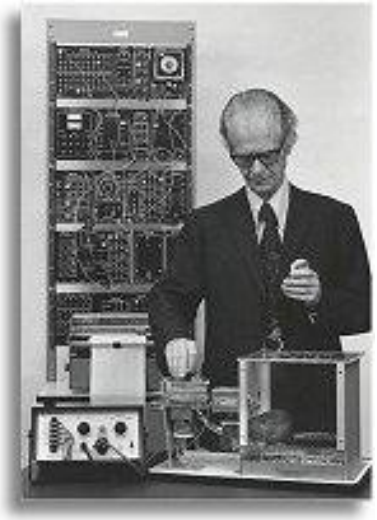


From ethology to behaviorism: animal learning behavior studies in controlled laboratory conditions



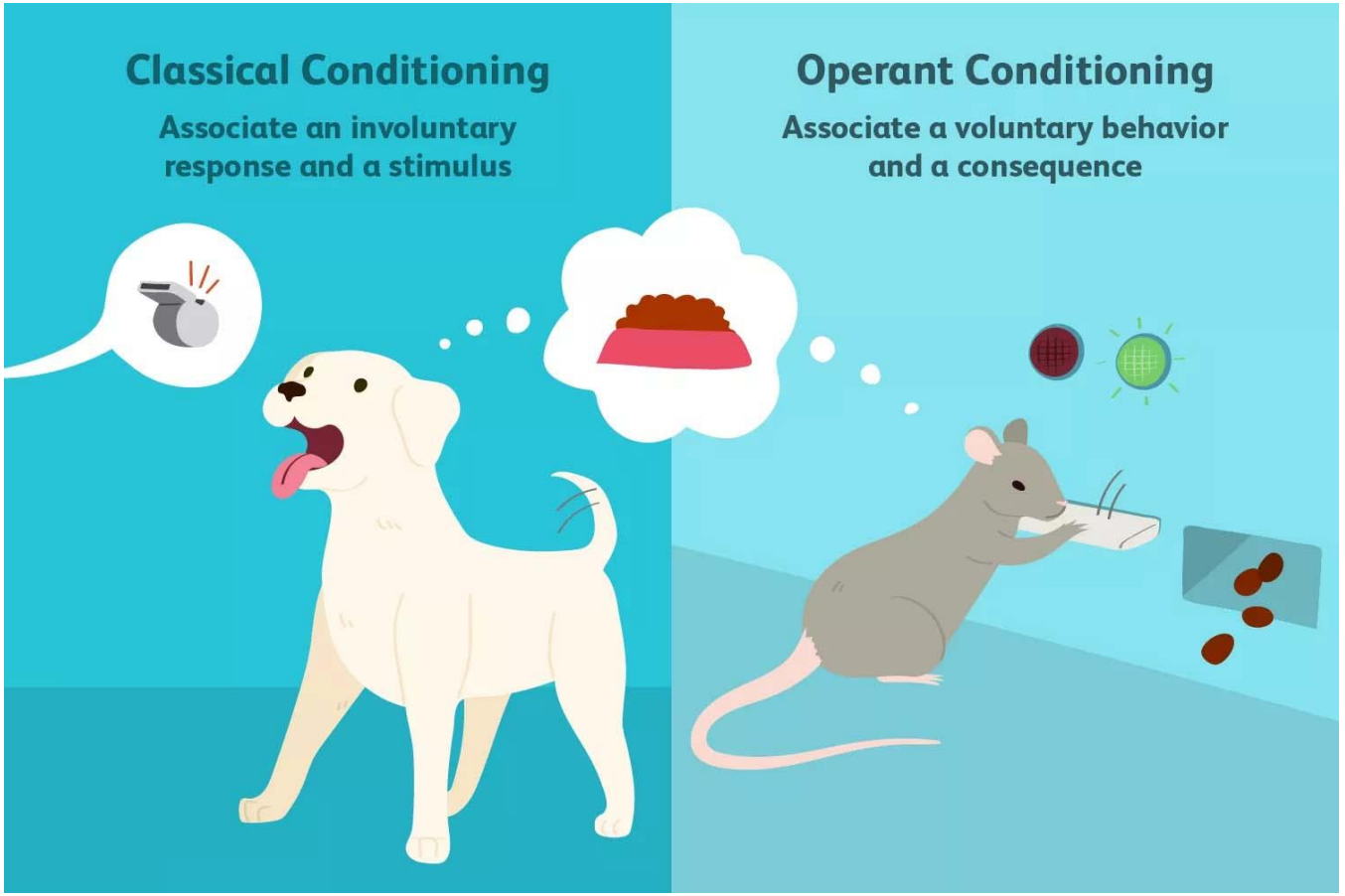


Ivan Pavlov
(1849-1936)

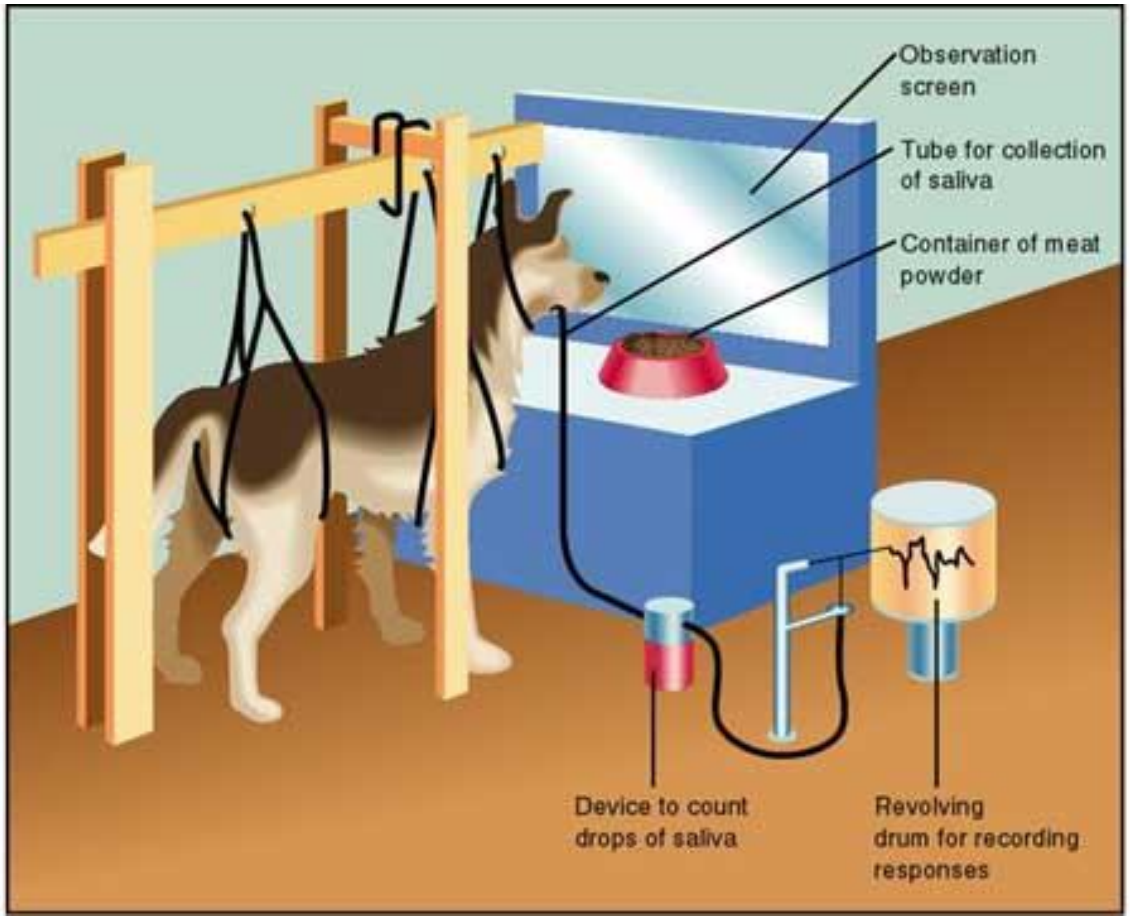


Burrhus Frederic Skinner
(1904-1990)

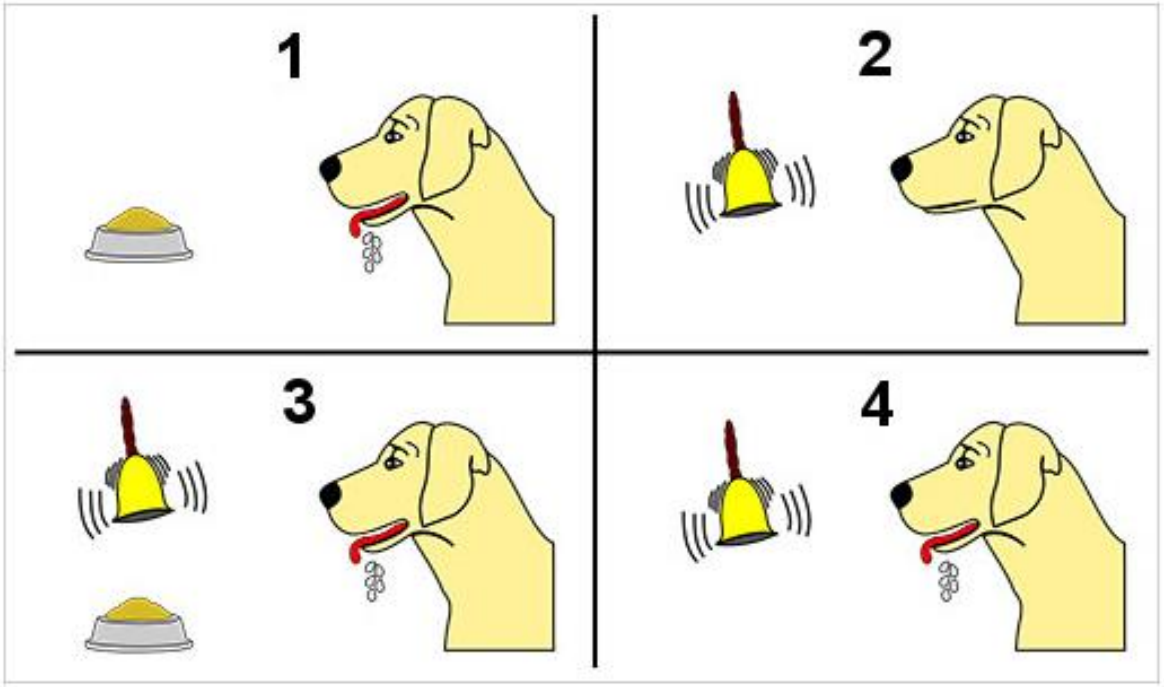
Pavlov **Skinner**

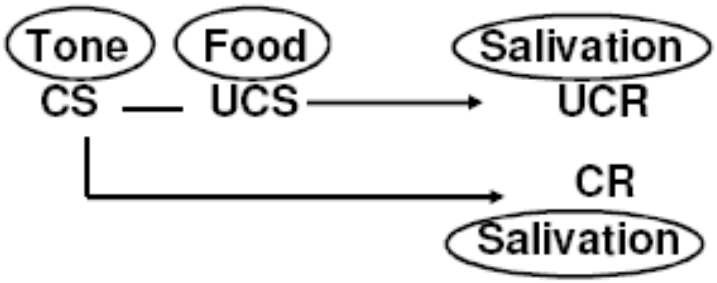


Pavlov's conditioning experimental setup

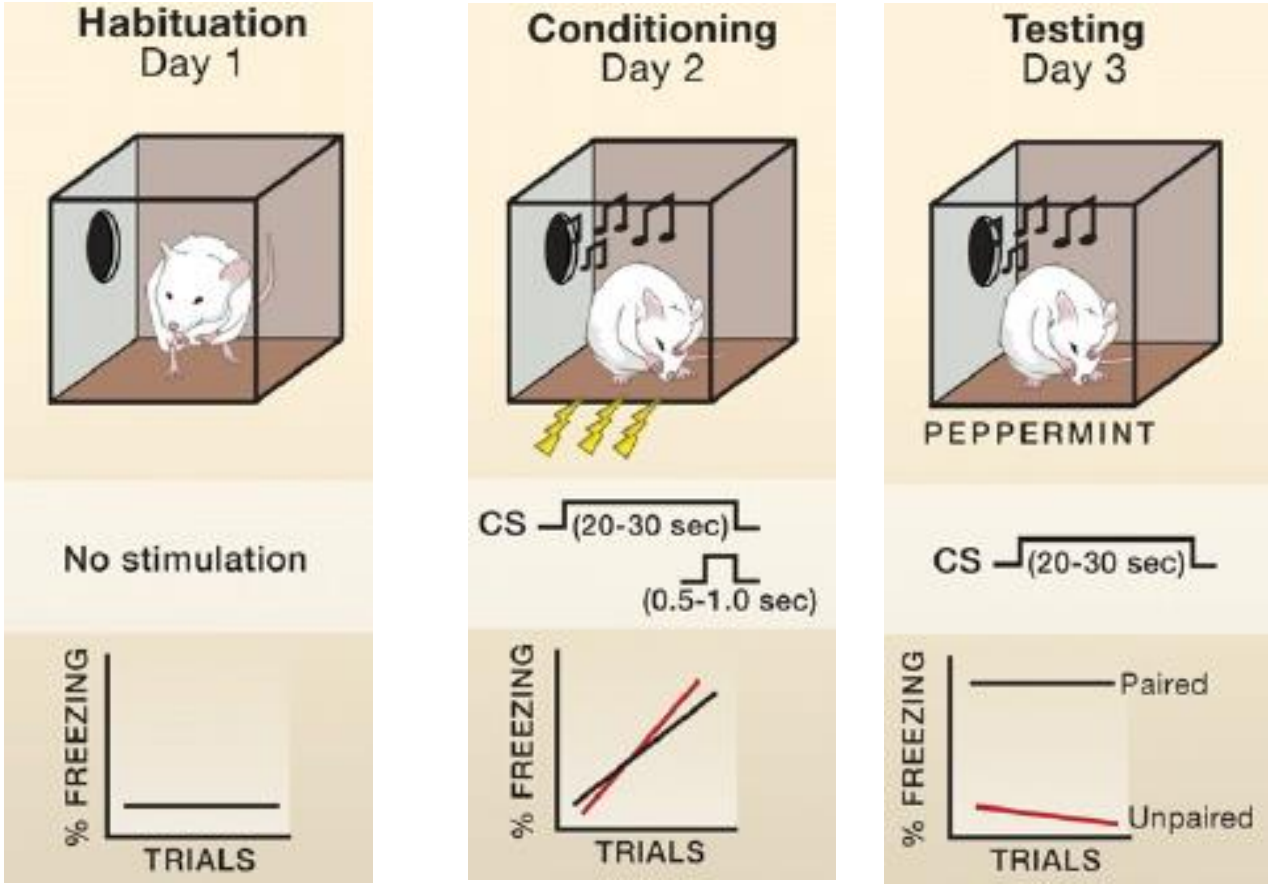


Pavlov's classical conditioning



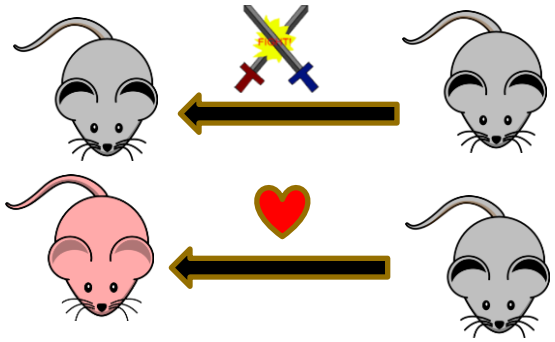


Classical conditioning in animal model



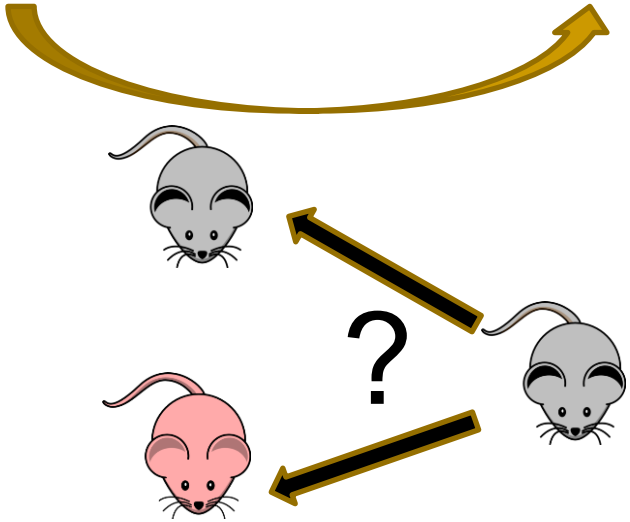
Jonathan et al 2011

Classical conditioning in innate social behavior



Conditional sex-specific pheromonal aversion

Associating female odor with a mild stomach ache /nausea



Conditioned taste aversion



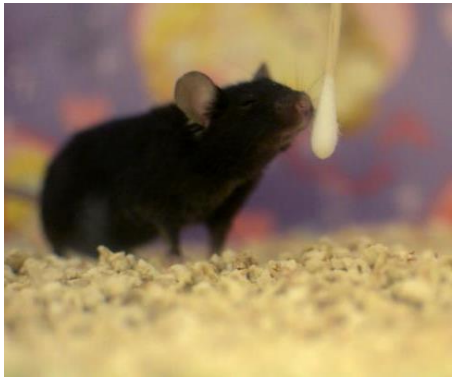


Transient sickness



Conditioned odor (pheromone) aversion

Female pheromones





Sexual behavior assay

Negative conditioning altered innate sexual behavior



Classical Conditioning in Human

Learn using positive/negative unconscious associations



1950



2026

Operant conditioning

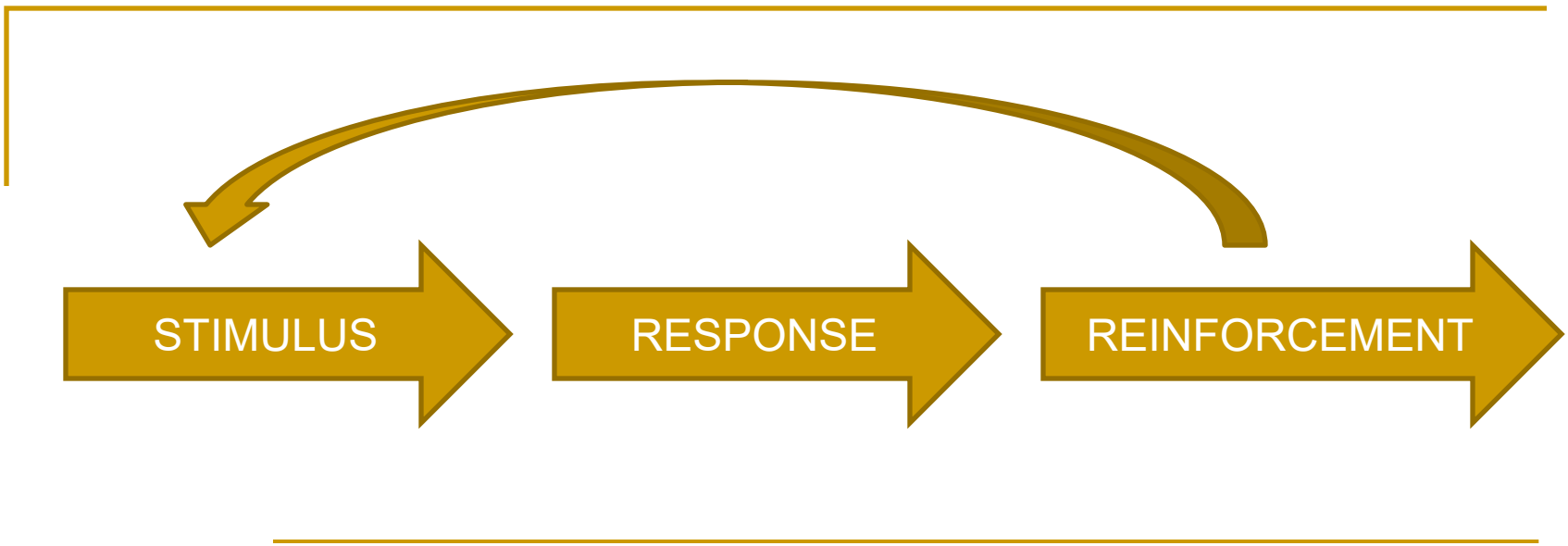


"Everything we do and are is determined by our history of rewards and punishments."-BF Skinner

- A process where an animal learns to associate one of its behaviors with a reward or punishment, and then tends to repeat or avoid that behavior

- In contrast to classical conditioning the response is voluntary (not a reflex) and the animal must do something to gain a reward (or avoid punishment)

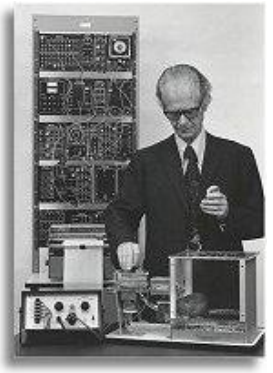
Operant conditioning

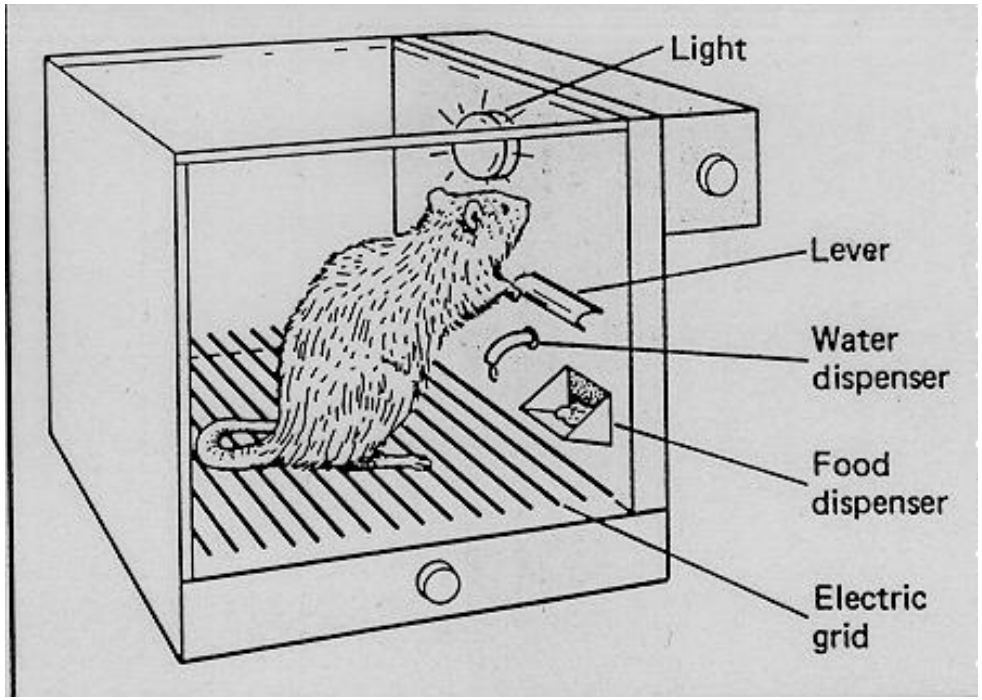


If your behavior is followed by a positive consequence, you are more likely to repeat the act in the future.

If it is followed by a negative consequence, you are less likely to repeat it.

Skinner's operant conditioning apparatus: Reinforcement or punishment





Positive reinforcement

Negative reinforcement

Operant conditioning in a rat



Operant conditioning in pigeons



Operant conditioning in human





Skinner Box

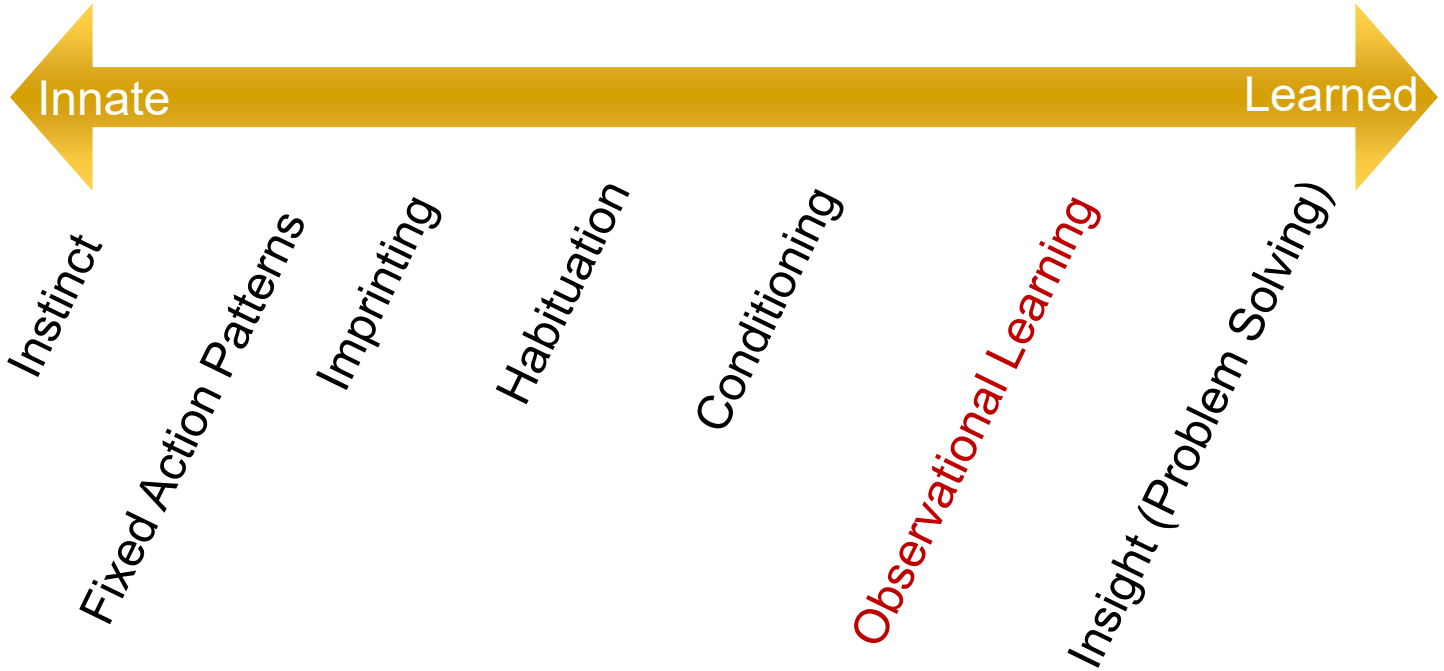


Slot Machine



Smartphone

Type of behaviors

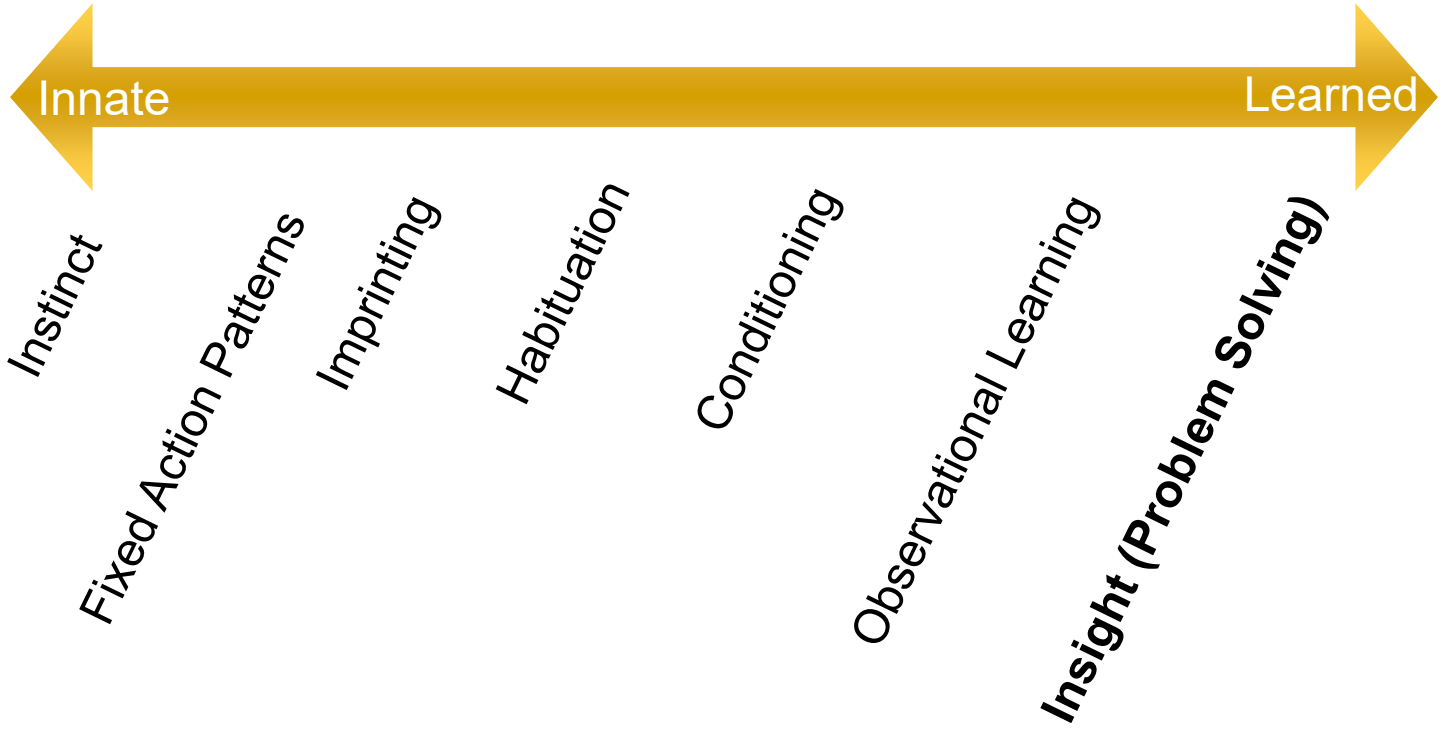


Observational Learning – Cultural transmission



Some Japanese monkeys learned to wash food before eating in 1950s, the behavior still persists in the group

Type of behaviors



Insight (problem solving)- Chimpanzees



Insight – Chimpanzee build tools



Jane Goodall



Insight – Caledonian crow tool building in nature



Insight – Weaver ants



Weaver ants use silk from larvae, and cooperatively build nests





Break

From Ethology to Neuroethology

A discipline that combines the study of natural animal behavior and neuroscience, to inquire how the brain controls behavior in wild animals, using lab conditions mimicking the natural environment or in the field.

The neural basis for seismic social communication

The blind mole rat (*Spalax*)

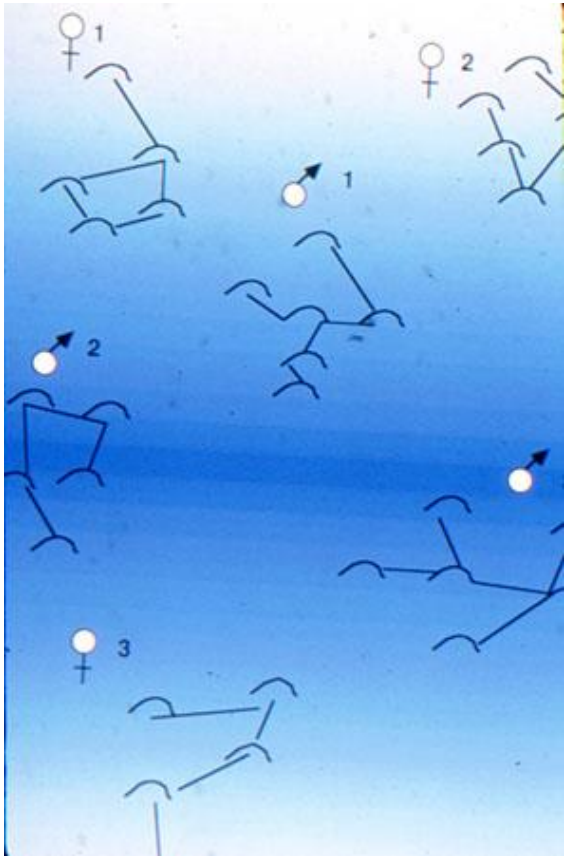


Sensory adaptation to the underground niche



How do blind mole rats communicate with each other (find mates/ avoid aggressive)?







Behavioral observations:

Mole rats produce head drumming
Mole rats often press their lower jaw to the tunnel side

Hypothesis:

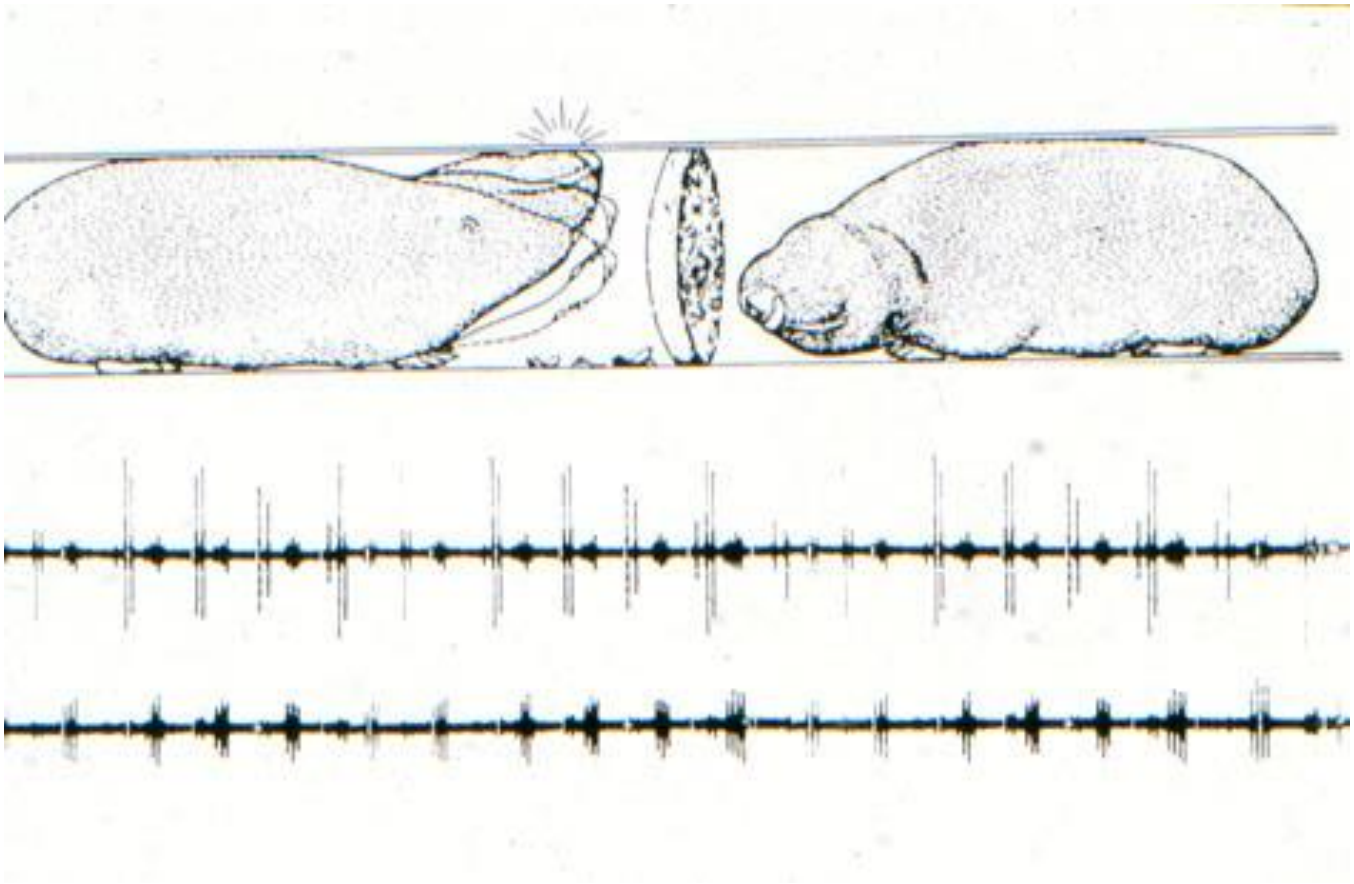
Mole rats communicate using soil-borne vibrations



Seismic social communication in blind mole rat



Social communication in the blind mole rat

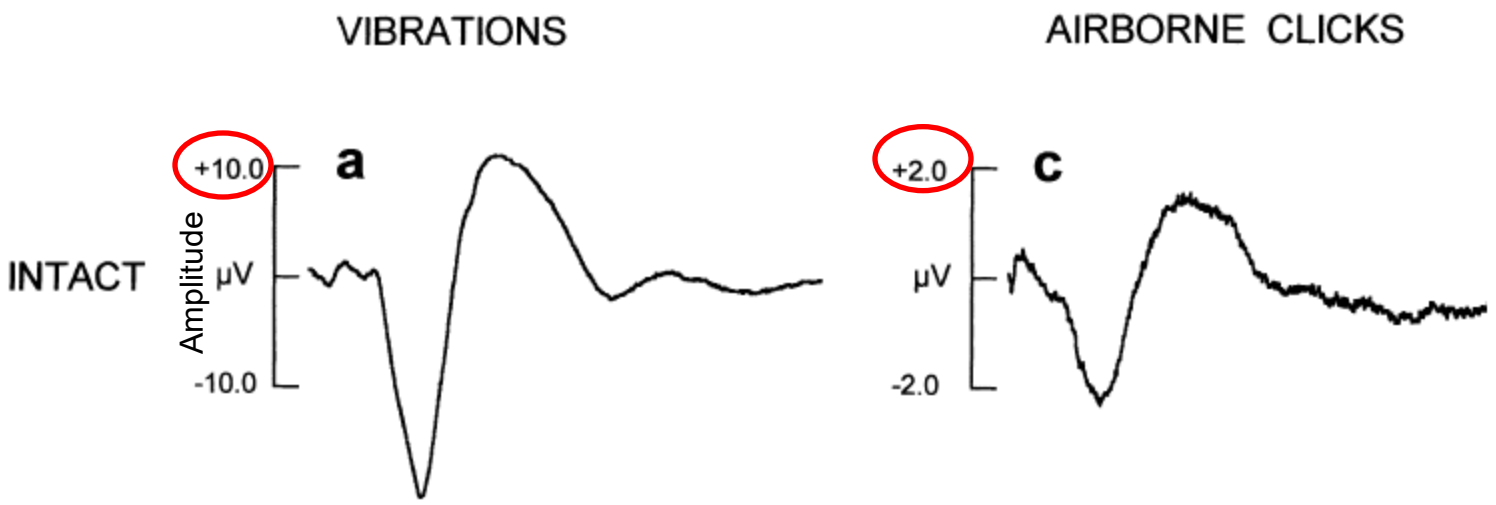


Lab experiments in the blind mole rat

Is head drumming perceived by the mole rat as air borne sounds or seismic signals?

Rado, R (1998); J Comp Physiol

Recording the response of the auditory system (the MLR) in mole rat following auditory and seismic signals

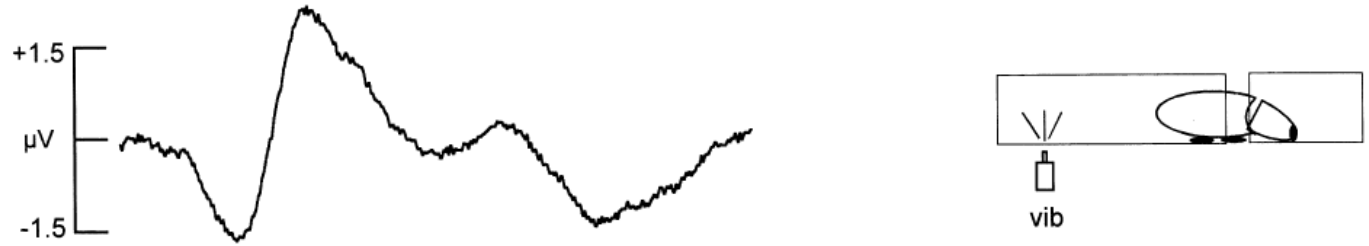


Middle latency response (MLR):
response of the auditory pathway occurring 20-70 ms after an auditory stimulus

Rado et al (1998); J Comp Physiol



a: Airborne sounds produced by the vibrator



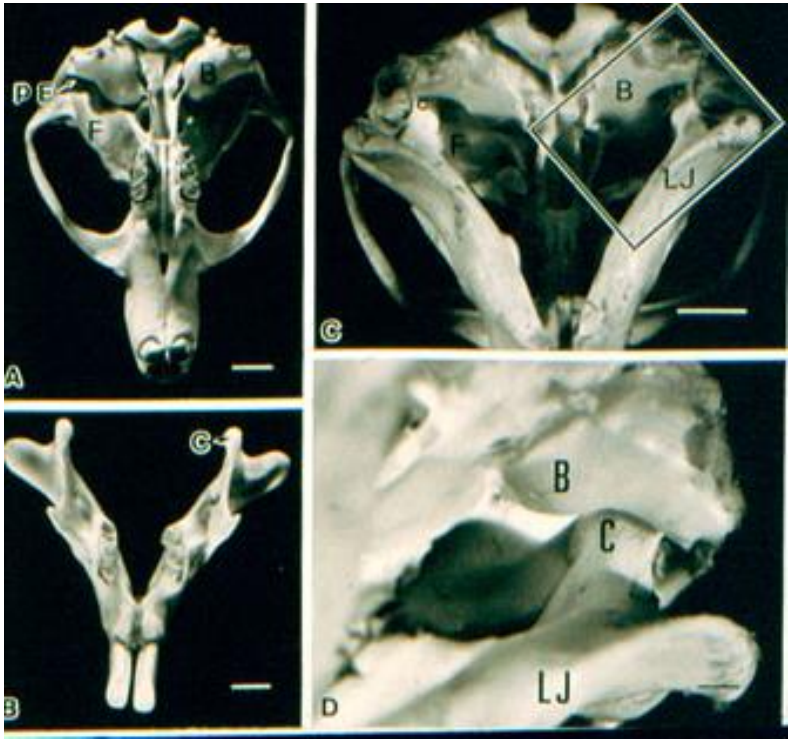
b: Vibratory stimulation of the entire body excluding head



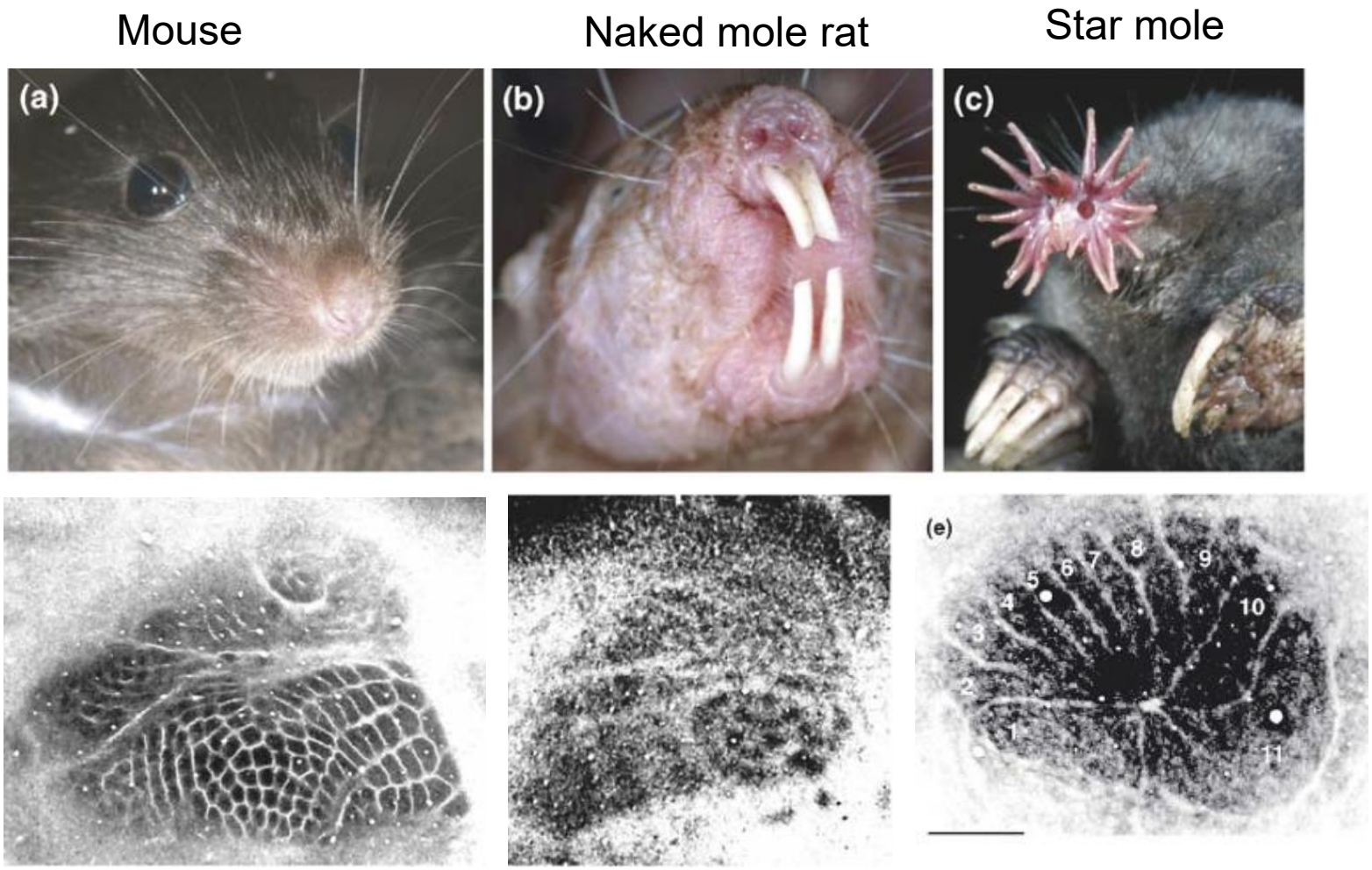
c: Vibratory stimulation of the head - mainly the lower jaw

Seismic signals are detected via bone conduction through the lower jaw





Species-specific somatosensory cortex map

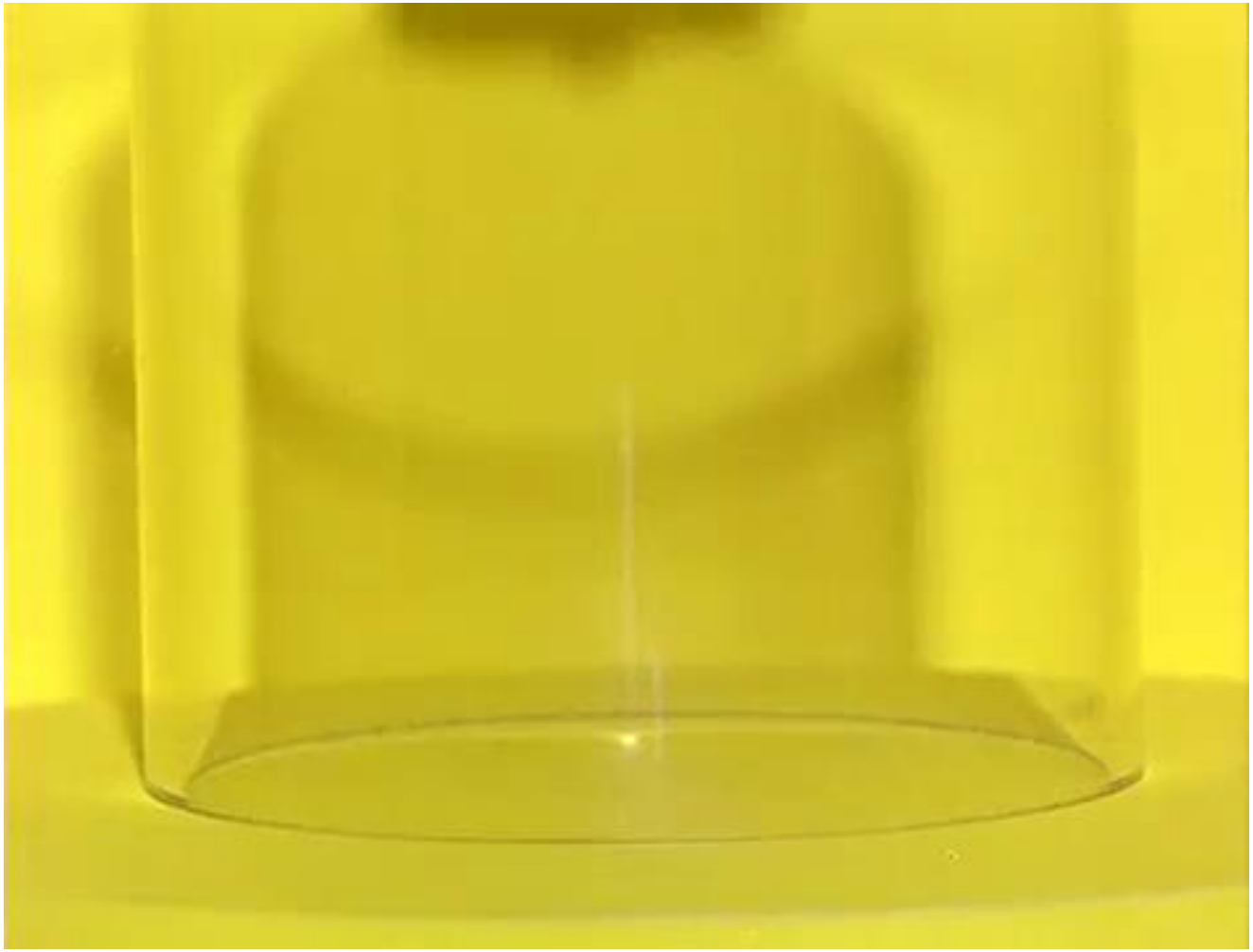


Catania and Henry 2006, Curr Opin Neurobiol

Neuroethology – Toad hunting worms



Neuroethology – Toad responds to horizontal moving segment



Neuroethology – Neuronal response in the toad brain



Neuron in the optic tectum (homologous to superior colliculus)

The neuronal basis of pair bonding in voles



Pair bonding and social behavior in voles

Prairie voles

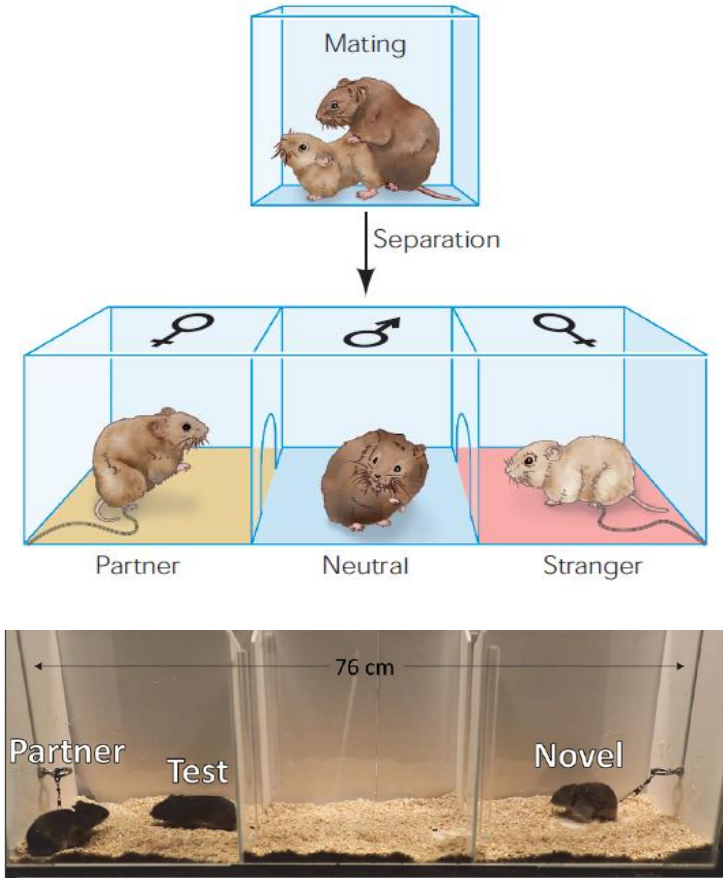
- Highly social
- Monogamous
- Spend most of their time in social interaction

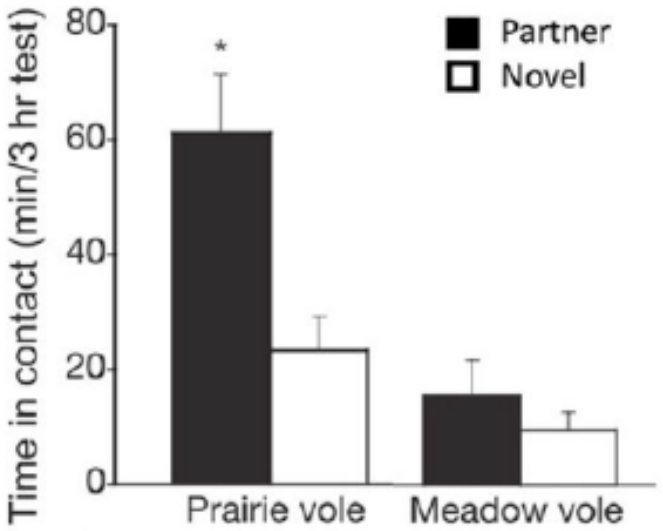


Montane/Meadow voles

- Avoid social contact except for the purpose of mating
- Polygamous
- Spend ~5% of their time in social interaction







Hemanth et al 2006

Sadino and Donaldson, 2018

Diversity in V1aR expression across species



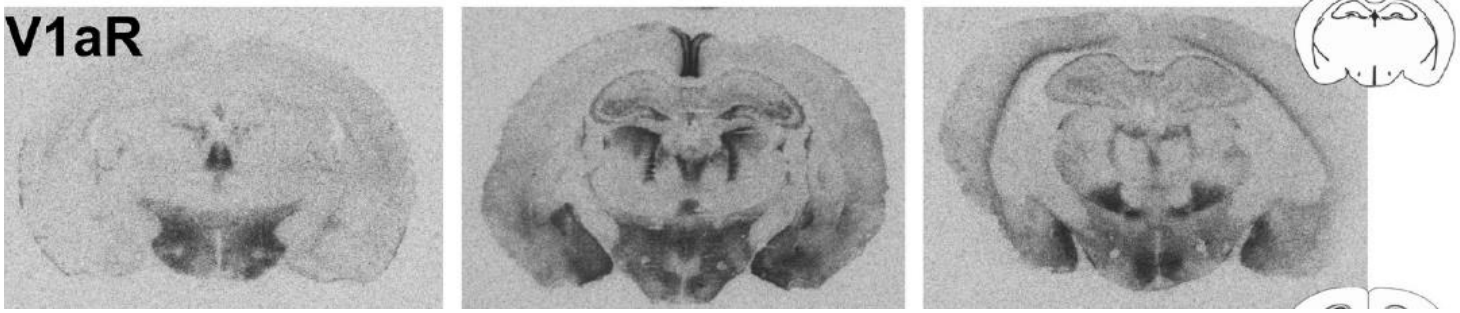




Mouse (Mus)

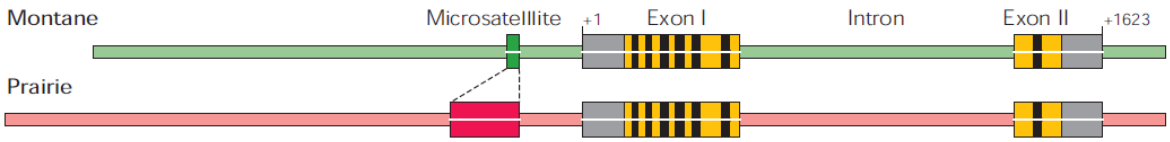
Prairie vole

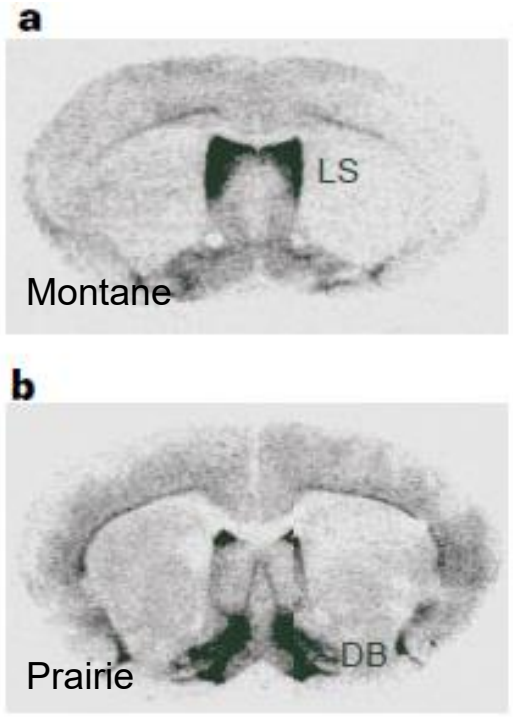
Meadow vole

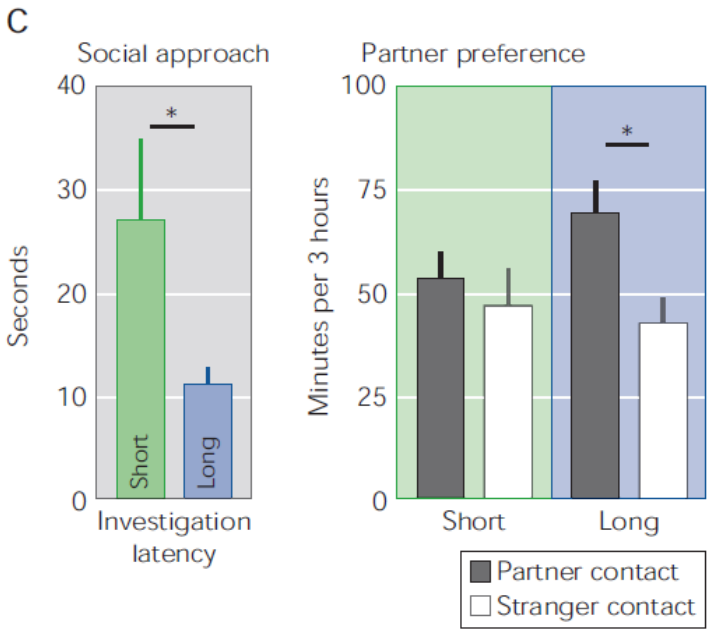


Sadino and Donaldson, 2018

Polymorphism in V1aR microsatellites generate differences in the brain and in social behavior







Hammock & Young, 2005 *Science*

Key takeaways: **Behavior is the central output of the brain and the entry point to neuroscience**

1. The brain exists to generate behavior

•Behavior is the *output* of neural computation

2. Behavior is shaped by evolution

•Natural selection → survival

•Sexual selection → reproduction

3. Behavior should be explained at multiple levels (Tinbergen)

•Mechanism (how?); Development (how over time?)

•Function (why?); Evolution (how did it arise?)

4. Behavior = interaction of innate programs and learning

•Hardwired circuits + plasticity

5. The brain constructs a species-specific world (Umwelt)

•Perception is not objective**.**

To understand the brain, study behavior in context

- Natural environment
- Appropriate model system
- Appropriate paradigms adapted to animal's umwelt

